

Forty-eight hours of an AI agent on the Kourovka Notebook: five solved problems, several partial results, and a record of the process

Claude Fable 5.1 (Anthropic)*

Josef Urban[†]

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Abstract

We report on a 48-hour experiment in which a general-purpose AI agent (Claude Fable 5.1 in the Claude Code harness, with GAP 4.16.1 and a 20-core budget) was asked to solve as many unsolved problems as possible from the 21st edition of the Kourovka Notebook and to document its work rigorously. The agent triaged the roughly 2300 open problems, launched about 190 GAP computations, and combined computer searches with short proofs. It produced complete answers to five problems (21.32: the property “ $G \cong H'$ for some finite H ” is decidable, with a quasi-polynomial witness bound; 16.60: $T(G) \geq |S_\alpha|$ for every automorphism α ; 2.78: the sets of soluble and of absolutely simple group-theoretic numbers are finite (the main question of 2.78); 18.46: Klyachko’s bound $|H| \leq 2|G|^2$ for square-root overgroups is sharp; 13.19: Gorchakov’s question on subdirect products has a negative answer for every prime), substantial partial results on seven further problems (16.14, 20.37, 20.21, 19.56, 21.26, 21.111, 18.117), and bounded-range verifications for about forty-five more. All proofs are self-contained and all computations are reproducible from a git repository. We describe the setting, the workflow, the mistakes made along the way, and what the experience suggests about using such agents for research mathematics. The status of the results with respect to the literature was not independently checked by the agent; some of the arguments may reproduce or vary known results.

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*The mathematics, the computations and this text were produced by the AI system Claude Fable 5.1 run inside the Claude Code agent harness; see Section 2. The human experimenter designed the task, set the constraints and supervised the run.

[†]Experiment design and supervision. Authorship and attribution conventions for AI-produced mathematics are unsettled; the reader should regard the AI system as the author of the content and the human as the experimenter.

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1 Introduction

The Kourovka Notebook [1] is a collection of unsolved problems in group theory that has been maintained since 1965; its 21st edition (2026) contains about 2300 problems that were open when the edition went to press, together with the identity of the proposers and comments on partial progress. It is a natural benchmark for the question that motivated the experiment reported here: *what can a current general-purpose AI agent, given a computer algebra system and two days, do with a large collection of genuinely open research problems?*

The experiment was deliberately simple in design. The agent received one instruction (reproduced verbatim in Section 2.1), a machine with GAP 4.16.1 installed, and a time limit of 48 hours; it was asked to document everything in a local git repository and warned that its solutions would be reviewed independently. The human experimenter did not intervene except to restart the agent’s goal loop after the initial hour and, at the very end, to ask for this write-up. Everything below—the choice of problems, the computations, the proofs, the reports, and this paper—was produced by the agent.

Outcome in one paragraph. The agent read the whole notebook (partly through delegated sub-agents), selected problems that looked attackable by finite computation or by a short argument, and worked on about sixty of them. Five problems received complete answers with self-contained proofs (Section 4, Appendix A):

- **21.32** (P. J. Cameron): it is decidable whether a finite group G is the derived subgroup of some finite group; if it is, a witness of order at most $|\text{Aut } G| \cdot |Z(G)|^2 \cdot (2 \exp Z(G))^{2 \lceil \log_2 |G| \rceil}$ exists (a quasi-polynomial bound), so the problem is in nondeterministic quasi-polynomial time.
- **16.60** (D. MacHale): $T(G) = \sum_{\chi \in \text{Irr}(G)} \chi(1) \geq |\{g : \alpha(g) = g^{-1}\}|$ for every automorphism α of every finite group G .
- **2.78** (P. I. Trofimov; main question): the sets of “soluble” and of “absolutely simple” group-theoretic numbers are finite; every $k \geq 86$ is the number of $\text{IE}\bar{n}$ -systems of a non-soluble non-simple group. (The second question of 2.78 and the related Problem 3.57, on the distribution of the non-soluble and simple numbers, remain open.)
- **18.46** (Ant. A. Klyachko): the bound $|H| \leq 2|G|^2$ for a group $H \supseteq G$ in which every element of G is a square is sharp: $G = D_8$ needs $|H| = 128$.
- **13.19** (Yu. M. Gorchakov): the answer is negative for every prime p ; in fact every finite p -group occurs as the quotient Q/H in the configuration of the problem.

Seven further problems received substantial partial results (Section 5), among them a structure theorem and a large exclusion range for Berkovich’s Problem 16.14, a general theorem and 267 explicitly realised factor sizes (286 verified factorisations) for Hooshmand’s Problem 20.37, and a proof that every minimal simple group violates the hypothesis of Monakhov’s Problem 19.56. About forty-five more problems were verified in bounded ranges without finding a counterexample (Section 6).

What this paper is and is not. This is a report on an experiment, written by the system that performed it. It records the setting, the process (including the mistakes), the mathematical content, and an assessment. It is *not* a claim of priority: the agent had no reliable way to search the literature during the run (web access existed but paywalled sources could not be read), and it is entirely possible that some of the “solved” problems have known answers, that some proofs reproduce known arguments, or that a problem’s status changed after the notebook was compiled. Where the agent knowingly reused a known theorem (for instance the twisted Frobenius–Schur theorem of Kawanaka and Matsuyama in Problem 16.60, or Zenkov’s theorem on minimal intersections in Problem 21.26) this is stated. We ask readers who know the literature to treat the results as candidates for verification, and we have tried to make verification as easy as possible: every proof is self-contained, every computation is a script in the repository, and every computational claim states exactly what was computed.

Organisation. Section 2 describes the setting (the task, the machine, the software, the agent). Section 3 describes how the run proceeded. Sections 4–6 present the results; the complete proofs are in the appendices, where they can be checked independently of the narrative. Section 7

discusses reliability, limitations and lessons. Appendix C gives the commit-level timeline and Appendix D the layout of the repository.

2 Setting

2.1 The task

The complete instruction given to the agent was:

Solve as many previously unsolved problems from THE KOUROVKA NOTEBOOK (docs/21tkr.pdf) as you can within an overall time limit of 48 hours. The gap system is installed in gap-4.16.1 but you can also download and install further SW if useful. In general, try to limit yourself to 20 CPU cores and 100G RAM on this machine while working. Write detailed plans and reports, iterate fast and create a local git repo keeping your work and documenting your progress, commit often there (but don't try to push). Work rigorously—your solutions will be independently reviewed.

The instruction was issued as a “goal” of the agent harness: a stop hook that re-invokes the agent whenever it tries to end its turn, so that the agent works continuously without further human input. After roughly one hour the experimenter re-issued the goal (“resume”), which also changed the reasoning-effort setting (see below). At the end of the 48 hours the experimenter asked for the present paper.

2.2 The material

The notebook was available as docs/21tkr.pdf together with a text extraction (docs/21tkr.txt, 15364 lines). The 21st edition lists the problems of all issues 1–21 that had not been moved to the archive of solved problems by the previous edition (2022); problems solved since then, or solved earlier without a published detailed solution, remain in the main part marked by an asterisk and commented (about 105 marks). The new problems of the 21st issue occupy lines 7885–9065 of the text file. None of the twelve problems treated in Sections 4–5 carries such a mark.

2.3 Hardware and software

The machine had 112 cores and 251 GB of memory; the agent was asked to stay within 20 cores and 100 GB, and did so approximately (the load average was mostly between 10 and 25, with short excursions to about 35 when several sweeps were launched at once). Software used:

- GAP 4.16.1 [8] with its standard package distribution, in particular the SmallGroups library (all groups of order ≤ 2000 except 1024), the libraries of transitive (degree ≤ 47 , no degree 32 data) and primitive (degree ≤ 4095) groups, the perfect groups library, CTblLib (character tables), AtlasRep (permutation representations of sporadic groups), ANUPQ (p -quotient and p -covering groups), Digraphs/bliss, irredsol, and the polycyclic package.
- nauty 2.8.6 [9] (geng, nauty.a), built from the copy shipped with GAP's Grape package, for graph enumeration (Problem 20.71).
- Python 3.12 in a virtual environment with networkx, python-sat (CaDiCaL) and scipy.
- Small C programs written during the run (a dancing-links exact-cover solver, permutation-group search loops, graph invariants).

2.4 The agent

The agent was Claude Fable 5.1 (Anthropic), run in the Claude Code harness (version 2.1.270) on the machine itself, with permission to execute shell commands without confirmation. The reasoning-effort setting was “xhigh” for the first hour and “high” afterwards. The harness provides a shell tool, a background-monitor tool (which turns lines of a log file into notifications), sub-agents (fresh copies of the model with their own context), web search and fetch, and automatic context compaction: when the conversation grows too long, the harness replaces the older part by a summary. One such compaction happened during the run (after about six and a half hours); the agent continued from the summary and from the files on disk.

Over the 48 hours the main agent produced 1232 assistant messages and issued 565 shell commands, 32 background monitors and 6 sub-agent calls (all six for reading and triaging parts of the notebook); the shell commands contain 193 invocations of GAP and about 180 background launches (`nohup`) of long computations. Sub-agents ran on the same model. The transcript (about 5400 JSON lines, 18 MB including sub-agent transcripts) is preserved in the `session/` directory of the repository.

2.5 Rules the agent set for itself

Early in the run the agent wrote a plan (`plans/PLAN.md`) with the following working rules, which it then followed: (1) triage the whole notebook for problems attackable by finite search or by a short argument; (2) launch cheap exhaustive searches immediately and keep them running in the background; (3) for anything that yields a proof or a counterexample, verify independently (a second script or a second method) and write a report with reproducible scripts; (4) for searches that find nothing, record the verified range as a partial result; (5) at most 20 GAP processes at a time, each limited to 8–16 GB.

3 How the run proceeded

3.1 Timeline

All times are UTC; the run started at 08:03 on 13 September 2026. Appendix C lists all 94 commits made during the run.

Hour 0–1 (triage). The agent created the repository, extracted the text of the notebook, read the new problems of the 21st issue itself (about 140 problems) and delegated the reading of issues 1–20 to five sub-agents, each covering about 1500 lines of text and asked to return a compact list of problems “settleable by a finite computation or a short rigorous argument”. Two of the sub-agents exceeded their output limit and had to be re-run with a request for a shorter report; part of one report (issues 19–20) was lost and this range was re-triaged about seven hours later. In parallel the agent wrote the first search scripts (Problems 21.113, 21.29, 21.99, 21.52, 21.26, 21.115) and launched them in the background with log monitors.

Hour 1–6 (first results). The first theoretical result, the decidability of Problem 21.32, was written and committed with its proof at 08:52; the solution of 16.60 followed at 09:13. The agent then worked through the candidates from the triage: for each it wrote a GAP sweep over the SmallGroups library (or over simple groups, primitive groups, character tables), started it, and moved on. By 13:00 about twenty sweeps were running. The solutions of 2.78 (13:26), 18.46 (13:52) and 13.19 (14:21, generalised at 14:38) were found in this period, each triggered by data from a sweep: for 18.46 the computation of minimal square-root overgroups showed that D_8 needs order 128; for 13.19 a search for the smallest configuration failed but suggested the class-2 construction that the agent then wrote down and proved.

Hour 6–7 (compaction). At about 14:40 the context of the main agent was compacted. The summary preserved the state of the running computations and the list of open threads, and the agent continued from it. A side effect surfaced at this point: the timestamps the agent had been writing into its progress log since about 13:00 were wrong by up to five hours (it had extrapolated the time instead of reading the clock). The error was detected by comparing with the git history and corrected.

Hour 7–12 (the two big problems). Problem 21.26 (Lisi–Sabatini) was verified up to order 2000 with a new double-coset checker. Problem 16.14 (Berkovich) was attacked theoretically: the agent proved a structure theorem for a minimal counterexample, realised that a counterexample of order 2^{11} with $r = 3$ would be a central extension of a group H of order 256 with $d(H) = 7$ satisfying a cohomological condition, and excluded all ten such H ; it then extended this to orders 2^{12} ($r = 3$), 2^9 and 2^{10} ($r = 2$). Problem 20.37 (Hooshmand) was started at 15:50: the agent computed which factorisations of simple groups are obtainable from subgroup chains, found that $\text{PSL}(2, 8)$ is the first group where this fails, proved a general theorem for the case of two cosets (a perfect-matching argument in vertex-transitive graphs), and designed an exact-cover search for the remaining cases. Problem 19.56 (Monakhov) was solved for minimal simple groups at 17:46, and 20.21 (Verret–Conder) received two impossibility lemmas.

Hour 12–40 (large sweeps). The second half of the run was dominated by long computations: the order- 2^{11} case of 16.14 required testing 7.1 million candidate quotients (about 6 hours on 14 cores, using the p -covering group instead of cocycle computations, an idea found at hour 12); Problem 20.21 was searched through all groups of order ≤ 1000 divisible by four (except two orders) and through the rank- ≤ 4 groups of order 512; the 20.37 exact-cover searches ran in several phases with increasing effort. Meanwhile the agent reviewed all reports line by line and corrected the wording of several claims.

Hour 40–48 and the end. The agent planned the final consolidation for three hours before the deadline but misread a 12-hour clock display and performed it at 09:44, 1 h 41 min after the deadline. No mathematical content was produced after the deadline: the final commit regenerated a table from data already on disk, re-verified all 20.37 factorisations in GAP, stopped the remaining processes and added a closing entry to the progress log.

3.2 The working pattern

Almost every problem was approached in the same way.

1. *Formalise the statement as a finite check* (a GAP function `Check(G)` returning a certificate of failure), test it on a few groups where the answer is known, and launch a sweep over a library in the background, usually split into several processes by order or by id range.
2. *Monitor* the log for hits and completions through the harness’s monitor tool rather than by polling.
3. *If a hit appears*, verify it with a second, independent computation (for 18.46 a lattice-based check instead of `IsomorphicSubgroups`; for 13.19 a pc-presentation of the constructed group; for 20.37 a direct multiplication of the sets in GAP after a Python search), then look for the general pattern behind it.
4. *If no hit appears*, record the exact range verified, and think about whether the data suggests a proof (as for 16.14, where the sweep showed that all inclusion-minimal Sylow intersections equal $O_p(G)$, or for 20.37, where the data showed which factor sizes need non-subgroup constructions).
5. *Write the report as soon as a result is stable*, commit, and update the index.

The theoretical work was interleaved with waiting for computations. The agent’s own account of its reasoning (preserved in the transcript) shows the typical pattern: a first idea, a concrete small case computed in GAP, a revision of the idea, and finally a proof written directly into the report.

3.3 Mistakes and how they were caught

A record of the run would be incomplete without its errors; all of them were caught by the agent itself.

- *Vacuous first version of a search* (20.122): the first sweep tested triples of Sylow subgroups only, for which the question is trivial by Zenkov’s theorem; the agent noticed this while writing the report and replaced the sweep by one over all nilpotent subgroups (which is much more expensive, and was therefore only completed for small orders).
- *Too strong a necessary condition* (20.21): the local condition was first formulated with an automorphism of order exactly 3; the correct condition needs an automorphism of 3-power order, and the search was repeated with the corrected condition.
- *A wrong side in a coset computation* (20.37): the first factorisation search used left cosets where right cosets were required; the verification step (direct multiplication) rejected the “solutions” and the bug was found immediately. This is the clearest example of the value of independent verification.
- *Clock drift in the progress log*, corrected from the git history (see above), and the late final consolidation caused by a 12-hour clock display.
- *Operational accidents*: a process-killing command that matched its own shell (three times), duplicated launches of the same sweep writing to the same log, GAP read-only global names, and a subagent report that exceeded the output limit. None affected the results.
- *Over-optimistic literature claims*: the agent initially cited “Konvisser’s classification” of 2-groups with three involutions as implying $d(G) \leq 4$; it could not access the paper and later marked this as an unverified literature dependency (it is now only needed for orders $\geq 2^{12}$ in Problem 16.14).

3.4 Resource use

The agent kept a running list of its processes and repeatedly stopped low-value sweeps to stay near the 20-core budget (the load average was above 30 for short periods only). Memory was never an issue (peak use well below 20 GB). The largest single computations were: the order- 2^{11} analysis for 16.14 (7.1 million groups, 6 hours on 14 cores); the order-512 rank- ≤ 4 search for 20.21 (420 514 groups, about 13 hours of wall-clock time on up to 8 cores, some blocks having been paused); the enumeration of all 1 006 700 565 connected graphs on 11 vertices for 20.71 (nauty, about 3 hours on 8 cores); and the 21.26 sweeps (about 800 000 groups). The repository (without the session transcript and the GAP installation) is about 530 MB, mostly logs and intermediate data.

4 Solved problems

For each of the five problems we state the problem as it appears in the notebook, the answer, and the idea of the proof; the complete proofs are in Appendix A. For each we also record what the agent knew about the literature status.

4.1 Problem 21.32 (P. J. Cameron): is “ $G \cong H'$ ” decidable?

Problem 1 (21.32). Is the following problem decidable, and if so, what is its complexity? Given a finite group G , is there a finite group H such that the derived subgroup of H is isomorphic to G ?

Theorem 4.1. *Let G be a finite group, $e = 2 \exp Z(G)$ and $m = \lceil \log_2 |G| \rceil$. If $G \cong H'$ for some finite group H , then $G \cong (H^*)'$ for a finite group H^* with $|H^*| \leq |\text{Aut } G| \cdot |Z(G)|^2 \cdot e^{2m}$. Consequently the problem is decidable, and it lies in nondeterministic quasi-polynomial time (a witness of order $|G|^{O(\log |G|)}$); no lower bound on the complexity is offered. Moreover, if $G \cong H'$ then some subgroup Q with $\text{Inn}(G) \leq Q \leq \text{Aut}(G)$ satisfies $Q' = \text{Inn}(G)$; when $Z(G) = 1$ this condition is also sufficient.*

Idea. Given H with $H' = G$, put $K = C_H(G)$. Then $H/K \leq \text{Aut } G$, $[H, K] \leq G \cap K = Z(G)$, and K has class ≤ 2 . Three reductions shrink H without changing H' : (1) replace H by the subgroup generated by G and $2m$ elements whose commutators generate G ; (2) the map $k \mapsto k^e$ is an endomorphism of K whose image C is central in H and of bounded index; (3) the central extension H of $\bar{H} = H/C$ by C can be replaced, using the universal coefficient theorem and a Baer-sum argument, by a central extension H^* of $\bar{H}$ by $C \cap G$ with the same derived subgroup. The last step is the only delicate one: one shows that the derived subgroup of a Baer sum $(E_1 \times_{\bar{H}} A)/\Delta$ is isomorphic to E'_1 when the Hopf map of A vanishes. The necessary condition follows by passing to $H/Z(G)$. *Literature.* The agent found no statement of this result; the problem was posed in the 21st issue (2026). Examples: S_3 and D_8 are not derived subgroups (known), $Q_8 = \text{SL}(2, 3)'$, $A_4 = S_4'$.

4.2 Problem 16.60 (D. MacHale): $T(G) \geq |S_\alpha|$

Problem 2 (16.60). For a finite group G let $T(G) = \sum_{\chi \in \text{Irr}(G)} \chi(1)$ and, for $\alpha \in \text{Aut}(G)$, $S_\alpha = \{g \in G : \alpha(g) = g^{-1}\}$. Is $T(G) \geq |S_\alpha|$ for all α ?

Theorem 4.2. *Yes. For every finite group G and $\alpha \in \text{Aut } G$, with $C = C_G(\alpha^2)$ and $\sigma = \alpha|_C$ (an involutory automorphism of C),*

$$|S_\alpha| = \sum_{\theta \in \text{Irr}(C)} \varepsilon_\sigma(\theta) \theta(1) \leq T(C) \leq T(G),$$

where $\varepsilon_\sigma(\theta) \in \{1, 0, -1\}$ is the twisted Frobenius–Schur indicator.

Idea. $S_\alpha \subseteq C_G(\alpha^2)$ and α restricts to an involutory automorphism of that subgroup; the twisted Frobenius–Schur theorem (Kawanaka–Matsuyama [2]) counts S_σ by indicators, and T is monotone on subgroups by Frobenius reciprocity. *Literature.* The two ingredients are classical; the reduction to the centraliser of α^2 is the only new step, and the agent regards this answer as “possibly known or easy for experts”. The classical case $\alpha = \text{id}$ is the inequality $T(G) \geq 1 + \#\{\text{involutions}\}$.

4.3 Problem 2.78 (P. I. Trofimov): group-theoretic numbers

Problem 3 (2.78). For a finite group G let $f(G)$ be the number of orders d such that G has a non-normal subgroup of order d (the number of $\text{IE}\bar{n}$ -systems). An integer k is soluble if every group with $f(G) = k$ is soluble, simple if $f(S) = k$ for some simple S (we read this as non-abelian simple; with abelian S admitted, 0 would be both soluble and absolutely simple), absolutely simple if in addition no non-soluble non-simple group has $f = k$, and composite if no simple group has $f = k$. Are the sets of soluble numbers and of absolutely simple numbers finite or infinite? Do composite non-soluble numbers exist?

Theorem 4.3. *Both sets are finite: every $k \geq 86$ equals $f(A_5 \times P)$ for a suitable p -group P ($p \geq 7$), which is non-soluble and not simple. Moreover $0, \dots, 6$ and 8 are soluble numbers, 7 is absolutely simple ($f(G) = 7$ iff $G \cong A_5$), and $f(S) \geq 9$ for every non-abelian simple $S \neq A_5$.*

Idea. A product formula $f(G \times H) = |\text{Ord}(G)| |\text{Ord}(H)| - n(G)n(H)$ for coprime orders, and an explicit family of p -groups $M_{p^n} \times C_{p^r}$ (modular group times cyclic) with $|\text{Ord}| = n + r + 1$ and $f = r + 1$, give $f(A_5 \times P) = 7n + 9r + 9$, which represents every integer ≥ 86 . The lower bound $f \geq 9$ for simple groups other than A_5 uses Thompson’s list of minimal simple groups and a count of subgroup orders; the case $f \in \{7, 8\}$ is settled by an analysis of minimal normal subgroups. *Literature.* The problem dates from the second issue (1965) and is still listed as open. The second question of 2.78 (do composite non-soluble numbers exist?) and Problem 3.57 (the distribution of non-soluble and simple numbers) are not answered; the data suggest that every $k \geq 9$ is a simple number.

4.4 Problem 18.46 (Ant. A. Klyachko): sharpness of $|H| \leq 2|G|^2$

Problem 4 (18.46). Every finite group G embeds in a finite group H in which every element of G is a square, and H can be chosen with $|H| \leq 2|G|^2$. Is this estimate sharp? (It cannot be improved beyond $|G|^2$.)

Theorem 4.4. *Yes. If $H \supseteq D \cong D_8$ and every element of D is a square in H , then $|H| \geq 128 = 2|D_8|^2$; the wreath product $D_8 \wr C_2$ attains the bound.*

Idea. Exhaustive verification: all groups of order $8, 16, \dots, 120$ are in the SmallGroups library, and two independent GAP computations show that none contains a dihedral subgroup of order 8 consisting of squares. Exactly 14 of the 2328 groups of order 128 do. The table of minimal $|H|$ for all groups of order ≤ 12 shows that D_8 is the smallest group needing the factor 2. *Literature.* The bound and the lower bound $|G|^2$ are from Baranov–Klyachko [10], cited in the notebook; the agent found no prior determination of the sharpness. Whether infinitely many groups need $2|G|^2$ was not investigated.

4.5 Problem 13.19 (Yu. M. Gorchakov): regularity of Q/H

Problem 5 (13.19). Suppose that a group Q and its normal subgroup H are subgroups of $G_1 \times \dots \times G_n$ such that for each i the projections of both Q and H onto G_i coincide with G_i . If Q/H is a p -group, is Q/H a regular p -group?

Theorem 4.5. *No, for every prime p . Every finite p -group P of class c occurs as Q/H in such a configuration with $n = c + 1$. The smallest example constructed has $|Q| = 2^9$, $n = 3$ and $Q/H \cong D_8$ (not regular, as regular 2-groups are abelian); no example exists with $|Q| \leq 192$.*

Idea. The reformulation is: Q has normal subgroups $N_1, \dots, N_n$ with $\bigcap N_i = 1$ and $HN_i = Q$ for all i (then $G_i = Q/N_i$). The class of Q/H is at most $n - 1$, so non-regular quotients need $n \geq 3$. The explicit example is a class-2 group of order 2^9 built from three copies of $\mathbb{F}_2^2$ and three central involutions; its properties were verified in GAP from a pc presentation. The general construction “tags” each generator of P by the subsets of $\{1, \dots, n\}$ containing a given index and uses a Möbius-inversion lemma on word maps in nilpotent groups to show that $\bigcap N_i = 1$. *Literature.* The problem is from the 13th issue (1995) and is listed as open; the class bound $\text{cl}(Q/H) \leq n - 1$ is elementary.

5 Partial results

The full statements and proofs of the theorems in this section are in Appendix B.

5.1 Problem 16.14 (Ya. G. Berkovich): 2-groups with central involutions

Problem 6 (16.14). Let G be a finite 2-group with $\Omega_1(G) \leq Z(G)$. Is $\text{rank}(G/G^2) \leq 2 \text{rank } Z(G)$?

Write $E = \Omega_1(G)$, $r = \text{rank } E = \text{rank } Z(G)$ and $d = d(G) = \text{rank}(G/G^2)$; the bound $d \leq 2r$ is attained by Q_8^r . The agent proved: (A) the answer is yes for groups of class ≤ 2 with $G^2 \leq Z(G)$, by a Chevalley–Warning argument applied to the squaring map $G/(Z(G)G^2) \rightarrow Z(G)/Z(G)^2$, which is a quadratic map without non-trivial zeros; (B) a counterexample of minimal order has $d = 2r + 1$, $E \leq \Phi(G)$, all maximal subgroups extremal with the same Frattini subgroup, every element $g \notin E$ has order $2|gE|$, every elementary abelian subgroup of $H = G/E$ has rank $\leq 2r$, every subgroup of H with trivial Schur multiplier is abelian, and $|G| \geq 2^{3r+2}$; (C) there is no counterexample of order $\leq 2^{11}$, and none of order 2^{12} with $r \geq 3$. Part (C) is a computation guided by (B): a counterexample G is a central extension of a group H in the SmallGroups library by $E = C_2^r$ in which no involution of H lifts to an involution; for each candidate H this is tested through the 2-covering group $F/R^2[F, R]$ of H (ANUPQ). All 7 111 878 groups of order 512 with $d(H) = 5$, the 1139 groups of order 512 with $d(H) = 7$, and all groups of order ≤ 256 with the relevant rank were excluded. The case $r = 2$, $|G| \geq 2^{12}$ is left open; it could presumably be closed with the classification of 2-groups with exactly three involutions [4, 5], which the agent could not consult.

5.2 Problem 20.37 (M. H. Hooshmand): subset factorisations $G = AB$

Problem 7 (20.37). Is it true that for every finite group G and every factorisation $|G| = ab$ there are subsets $A, B \subseteq G$ with $|A| = a$, $|B| = b$ and $G = AB$? (A minimal counterexample is simple without prime-index subgroups.)

Call a factor size a *chain-achievable* if it can be obtained from subgroup chains and transversals. The agent computed the chain-achievable sizes for all non-abelian simple groups of order $\leq 10^6$; $\text{PSL}(2, 8)$ is the smallest group with non-chain-achievable sizes ($12 \cdot 42$ and $21 \cdot 24$), and 43 of the 56 simple groups below 10^6 have such sizes—400 pairs in all. Two constructions cover most of them:

Theorem 5.1. *Let $H \leq K \leq G$ with $|K : H|$ even and $K = \langle H, x \rangle$ for some x . Then there is $B \subseteq K$ with $K = (H \cup Hx)B$, so $2|H|$ is a factor size of K and of G .*

The proof is a perfect matching in a connected vertex-transitive graph on the coset space $H \backslash K$, which exists by a classical consequence of the Gallai–Edmonds structure theorem [6]. The second construction takes $A = HX$ (a union of right cosets of H) and $B = YK$ (a union of left cosets of K) for subgroups H, K with regular double cosets; $G = AB$ then reduces to an exact cover of the double coset space $H \backslash G / K$, which is small and was solved by a randomised dancing-links search [11]. In this way 267 of the 400 sizes were realised by explicit sets, every one re-verified in GAP by direct multiplication; $\text{PSL}(2, q)$ is completely settled for $q \leq 47$ and many larger q . The smallest open cases are $70 \cdot 288$ and $140 \cdot 144$ for $\text{PSL}(3, 4)$ and $70 \cdot 416$, $140 \cdot 208$ for $\text{Sz}(8)$, which resisted $3 \cdot 10^5$ random attempts.

5.3 Problem 20.21 (G. Verret / M. Conder): $K \cong L$ with $G/K \cong C_{12}$, $G/L \cong A_4$

Problem 8 (20.21). Is there a finite group G with normal subgroups $K \cong L$ of index 12 such that $G/K \cong C_{12}$ and $G/L \cong A_4$?

The agent showed that in an example of minimal order $I = K \cap L$ satisfies $K/I \cong V_4$, $L/I \cong C_4$, $G/I \cong C_4 \times A_4$, and K must satisfy a “local condition” (L): K has normal subgroups I and $I' \cong I$ with $K/I \cong V_4$, $K/I' \cong C_4$, and an automorphism stabilising I that permutes the non-trivial cosets of I cyclically. Two lemmas exclude (L) for abelian K and whenever $I \cdot I' \neq K$.

(an involution-counting argument in two coset decompositions); computationally, no group of order ≤ 2000 is an example (direct search for orders divisible by 12 with $\leq 30\,000$ groups, and the exclusion of (L) for $|K| \leq 256$, which covers all orders ≤ 3072), and no group of order 512 and rank ≤ 4 , or of order 260–1000 divisible by 4 (except 512 of rank ≥ 5 and 768) satisfies (L). An example, if one exists, has $|G| \geq 6144$.

5.4 Problem 19.56 (V. S. Monakhov): $\star$ -commutators

Problem 9 (19.56). A $\star$ -commutator is a commutator $[x, y]$ of elements of coprime prime-power orders. Is G soluble if $|ab| \geq |a||b|$ for all $\star$ -commutators a, b of coprime orders?

Theorem 5.2. *Every minimal simple group contains an involution a and an element b of odd prime order, both $\star$ -commutators, with ab of order 2; hence it violates the hypothesis (M). Consequently every counterexample to 19.56 (a non-soluble group satisfying (M)) contains a minimal non-soluble subgroup which is again a counterexample and has non-trivial Frattini subgroup; in particular a counterexample of minimal order is minimal non-soluble with non-trivial Frattini subgroup.*

The proof is uniform over Thompson’s list except for $\text{PSL}(3, 3)$, which was done by computation: involutions are $\star$ -commutators via A_4 or a Borel subgroup, elements of odd prime order via a dihedral subgroup ($[b, t] = b^{-2}$). The 208 non-soluble groups of order ≤ 2000 (in orders with ≤ 5000 groups) and all 443 perfect groups of order $\leq 10^5$ (including many with large Frattini subgroups) violate the hypothesis; the agent could not handle Frattini extensions uniformly.

5.5 Problem 21.26 (F. Lisi, L. Sabatini): simultaneous minimal Sylow intersections

Problem 10 (21.26). Let $p_1, \dots, p_k$ be the prime divisors of $|G|$ and H_i Sylow p_i -subgroups. Is there $x \in G$ such that $H_i \cap H_i^x$ is inclusion-minimal in $\{H_i \cap H_i^g\}$ for all i simultaneously?

The answer is yes when $|G|$ has exactly two prime divisors (so that $G = PQ$ for the two Sylow subgroups): if $P \cap P^b$ and $Q \cap Q^{a'}$ are minimal with $b \in Q$, $a' \in P$, then $x = ba'$ works for both. The general case was verified for all groups of order ≤ 2000 (except four 2-heavy orders) and for about sixty larger groups (symmetric, linear, unitary, Suzuki, Mathieu, wreath products); in every case all inclusion-minimal intersections equal $O_p(G)$, as predicted by Zenkov’s theorem [3], and a common x was found within a few random samples.

5.6 Problem 21.111 (D. O. Revin, Zh. Yang): τ -groups

A non-identity automorphism x (inner or outer) of a simple group S is a τ -automorphism if any two conjugates of x in $\langle x, \text{Inn } S \rangle$ generate a $3'$ -group; S is a τ -group if it has one. The agent showed that $\text{PSL}(2, 3^n)$ is a τ -group for every $n \geq 2$ (an involution inverting no unipotent element), and verified that among the simple groups of order $\leq 10^7$ divisible by 3 exactly $\text{PSL}(2, 9)$, $\text{PSL}(2, 27)$, $\text{PSL}(2, 81)$ and $\text{PSL}(2, 243)$ are τ -groups, all τ -automorphisms found being involutions. This suggests the answers “ $\text{PSL}(2, 3^n)$, $n \geq 2$ ” to part (a) and “no” to part (b).

5.7 Problem 18.117 (P. Shumyatsky): commutators of coprime-order elements

For 15 sporadic groups (M_{11} , M_{12} , M_{22} , M_{23} , M_{24} , J_1 , J_2 , HS, J_3 , McL, He, Co_3 , Co_2 , Suz, Fi_{22}) every conjugacy class contains a commutator $[x, y]$ with $\gcd(|x|, |y|) = 1$; each class comes with an explicit certificate found by random or targeted search in a permutation representation.

6 Bounded-range verifications

For the problems below the agent found no counterexample in the stated range. Each entry names the directory of the repository with the scripts and logs. “Orders with $\leq N$ groups” means that orders whose SmallGroups count exceeds N were skipped; 512, 1024 and 1536 are always skipped unless stated. The table is the final version of `reports/verifications.md`.

Problem	Statement checked	Range verified
21.29 (Burness–Giudici)	primitive G with a regular suborbit: $\forall \alpha, \beta \exists \gamma$ with $G_{\alpha\gamma} = G_{\beta\gamma} = 1$	all 10 545 primitive groups of degree ≤ 4095 with a regular suborbit (only S_2 on 2 points fails, degenerately)
21.99 (Müller)	transitive G , $\alpha \neq \beta \Rightarrow \exists g$: $\alpha^g = \beta$, $ \text{fix } g \neq 1$	all transitive groups of degree ≤ 47 except 32
21.113 (Robinson)	$\Psi_{p,G}$ is a character; projective character	all groups of order ≤ 1000 except 512, 768; all perfect groups of order $\leq 10^5$; all primitive groups of degree ≤ 40 and order $\leq 2 \cdot 10^6$; 1663 library tables of order $\leq 10^7$ (Brauer tables where available); p -soluble groups ≤ 400
21.26 (Lisi–Sabatini)	simultaneous minimal Sylow intersections	all groups of order ≤ 2000 except 1024, 1280, 1536, 1792; ~ 60 larger groups (see Section 5)
21.52/21.53 (Gorshkov)	automorphisms of the $ ab $ -coloured complete graph on an involution class	all simple L of order ≤ 7920 , all involution classes
21.25 (Breuer–Guralnick)	simple $G = \langle P_1, P_2 \rangle$ for Sylow subgroups of any two primes	all 66 simple groups of order $\leq 2 \cdot 10^6$
21.27/21.28 (Burness et al.)	products of two derangements; $\delta(G) \geq 89/325$	all simple groups of order $\leq 5 \cdot 10^5$, all transitive actions; minimum δ observed 2/7
21.135/21.59(a)	character-degree multisets vs trivial radical / almost simple	the groups of orders 60, 120, 168, 336, 360, 504, 660, 720 with trivial radical
21.2 (Amiri)	order-divisibility bijection onto a simple group forces simplicity	orders 60, 168, 360, 504, 660, 1092
20.115 (Wilde)	$\chi(x) \neq 0 \Rightarrow o(x) \mid G /\chi(1)$	the 208 non-soluble groups ≤ 2000 (orders with ≤ 5000 groups); 2275 library tables with ≤ 300 classes
20.78 (Qian codegree conjecture)	element of order $m \Rightarrow m$ divides some codegree	non-soluble groups ≤ 2000 (orders with ≤ 5000 groups) with $F(G) \neq 1$; 2275 library tables; perfect groups $\leq 10^5$ with $F(G) \neq 1$
19.108 (Wilde)	p -groups, p odd: $\chi(x) \neq 0 \Rightarrow o(x) \mid P /\chi(1)^2$	all non-abelian groups of order 3^3 – 3^7 , 5^3 – 5^5 , 7^3 – 7^4
20.49 (Lucchini)	2-generated subgroup of the same exponent	the 208 non-soluble groups ≤ 2000 (orders with ≤ 5000 groups)
20.4(b),(c) (Bächle)	cut groups: Sylow 3-subgroups cut; $\exp O_p(G) = p$ ($p = 5, 7$)	all cut groups of order ≤ 2000 divisible by 3, 25 or 49 (orders with ≤ 5000 groups, except 512, 1024, 1536)
20.122 (Zenkov)	$\min_G(A, B, C)$, $\text{Min}_G(A, B, C) \leq F(G)$ for nilpotent A, B, C	all nilpotent subgroups, orders ≤ 95 and parts of 123–190
20.71(a),(b) (Lauri/Pannone)	connected graph with $k \in \{2, 3\}$ card types and $> k$ orbits	all connected graphs on ≤ 11 vertices (1 006 700 565 on 11 vertices)
20.86 (Schmid)	Gheri’s inequality for non- p -soluble groups	the 208 non-soluble groups ≤ 2000 (orders with ≤ 5000 groups); the 73 simple groups $\leq 3 \cdot 10^6$; all perfect groups $\leq 2 \cdot 10^5$
20.30 (Neumann–Vaughan–Lee)	perfect centreless G : $ G \leq n^2$ ($n = \max$ class size)	all perfect groups of order $< 10^6$ with trivial centre

Problem	Statement checked	Range verified
20.55 (Mattarei)	identical character tables, derived lengths 2 and 4	soluble groups ≤ 1000 (orders with ≤ 3000 groups)
20.99 (Sun, $k = 5$)	5 disjoint cosets with pairwise gcd of indices ≤ 4	groups of order ≤ 31
20.100 (Sun, $n = 4$)	4-subsets orderable with a_1, a_2^2, a_3^3, a_4^4 distinct	all 85 groups of order coprime to 30 and ≤ 300
19.12 (Bezerra et al.)	every finite simple group is conjugacy-expansive	34 simple groups with ≤ 24 classes (order $\leq 3 \cdot 10^6$)
19.20 (Cameron)	$ \text{End}(G) = \text{PIso}(G) \Rightarrow G$ abelian	all groups of order ≤ 120
19.30	same order and same vanishing-element orders as $S \Rightarrow G \cong S$	$S = A_5, \text{PSL}(2, 7), A_6, \text{PSL}(2, 8), \text{PSL}(2, 11), \text{PSL}(2, 13)$
19.33 (Giannelli)	χ_P with a linear constituent has $\geq p$ of them	non-soluble groups ≤ 2000 (orders with ≤ 5000 groups); simple groups $\leq 10^6$
18.20 (Belonogov)	semiproportional irreducibles have equal degree	all groups ≤ 250 ; 2063 library tables with ≤ 200 classes
18.44 (Keller)	only $(D_8, \mathbb{F}_3^2)$	non-abelian groups ≤ 120 , modules over $\mathbb{F}_q$, $q \leq 13, V \leq 2 \cdot 10^5$
18.56 (Cutolo)	core-2 2-groups have an abelian subgroup of index ≤ 4	all 2-groups of order ≤ 128
18.77 (Passman)	kernels of max-degree irreducibles of p -groups intersect trivially ($p > e$)	p -groups of order $2^4\text{--}2^8, 3^3\text{--}3^7, 5^3\text{--}5^5$
18.114 (Schmid)	$k(VG) = V $ for irreducible $5'$ -subgroups of $\text{GL}(n, 5)$	$n \leq 3$: only Singer cycles
18.119 (Shumyatsky)	Sylow subgroups of $w(G)$ generated by w -values	all groups ≤ 240 for the words $[x, y], [x, y, z], \delta_2, \delta_3, [[x, y], [z, t], u], [[x, y, z], [t, u]]$; δ_2 and γ_3 for all groups ≤ 2000 (orders with ≤ 5000 groups) of derived length ≥ 3 with non-nilpotent G''
18.120 (Jabara)	$P = AB$, A abelian, B class 2 $\Rightarrow \langle A, [B, B] \rangle = AB_0$	all p -groups of order $16\text{--}64, 27\text{--}243, 125$
17.1 (Isaacs)	partition by subgroups of equal order with a non-abelian component	exponent-3 groups of order 3^6 ; exponent-5 groups of order 5^6
17.128 (Jabara)	T of exponent p if all elements of $TA \setminus T$ have order p	3-groups of order $\leq 3^5$
16.95 (Thompson)	$\exists$ permutation P with AP cyclic	$\text{GL}(2, q)$ for $q \in \{2, 3, 5, 7\}$; $\text{GL}(3, q)$ for $q \in \{2, 3, 5\}$; $\text{GL}(4, 2), \text{GL}(4, 3), \text{GL}(5, 2)$
16.45 (Cameron)	$b(G) = \mu'(G)$	all non-cyclic groups of order ≤ 32
16.63/15.29	$ \text{Aut } G = G $ for p -groups; $\text{Aut } G \cong G$ for 2-groups	p -groups of orders $3^4\text{--}3^7, 5^3\text{--}5^6, 7^3\text{--}7^5, 11^3, 11^4, 13^3, 17^3$; 2-groups $\leq 2^7$
16.33 (Glauberman)	abelian B of maximal order normal in $\langle B^G \rangle$	all 2-groups of order ≤ 128
16.14 (Berkovich)	$d(G) \leq 2 \text{ rank } Z(G)$	all 2-groups of order $\leq 2^{11}$ (Section 5)
15.2(a) (Isaacs)	non-vanishing even-order elements of soluble G lie in $F(G)$	soluble groups of even order ≤ 1000 except 512, 768
15.3 (Isaacs)	soluble G with faithful non-linear α, β and $\alpha\beta$ irreducible	soluble groups ≤ 600 (orders with ≤ 3000 groups)
14.44/14.74	$k(AB) \leq k(A)k(B)$; $k(G) \leq \prod k(P_i)$	all groups of order ≤ 382
11.8(b) (Berkovich)	same degree multiset, one soluble $\Rightarrow$ other soluble	all orders ≤ 1200 with $\leq 20\,000$ groups
11.17(b) (Brandl)	$\exp(G)$ divides the Engel period for simple G	$\text{PSL}(3, 3), \text{PSU}(3, 3), M_{11}, \text{PSL}(3, 4), \text{PSU}(4, 2), \text{Sz}(8), \text{PSU}(3, 4), M_{12}, \text{PSL}(3, 5), \text{PSU}(3, 5), J_1$

Problem	Statement checked	Range verified
11.18/17.80 (Brandl, Macedoska)	$G(2, 3) = \langle x, y \mid x = [x, y, y], y = [y, x, x, x] \rangle$	perfect; no subgroup of index ≤ 12 ; no quotient onto a simple group of order $\leq 2 \cdot 10^5$; coset enumeration fails at $4 \cdot 10^8$ cosets
21.111	τ -groups	simple groups of order $\leq 10^7$ (Section 5)
18.117	coprime commutators	15 sporadic groups (Section 5)
19.56	$\star$ -commutator hypothesis	non-soluble groups ≤ 2000 (orders with ≤ 5000 groups); perfect groups $\leq 10^5$
20.21	existence of the configuration	groups ≤ 2000 ; local condition (L) excluded for $ K \leq 1000$ except $ K = 512$ of rank ≥ 5 and $ K = 768$ (Section 5)

Two of these deserve a comment. In 21.26 the sweep also recorded, for every group, that all inclusion-minimal intersections $P \cap P^x$ equal $O_p(G)$ —an instance of Zenkov’s theorem [3] that the agent used as a sanity check of the code. In 20.71 the computation is a nauty-based enumeration with a C filter and is the largest single enumeration of the run; the case $k = 4$ (already solved in the literature) served as a positive control, since the filter finds examples on 9 vertices for $k = 4$.

7 Discussion

7.1 What the results are worth

The five “solved” problems fall into three types. Two (18.46 and 13.19) are decided by an explicit object: a group of order 128 (respectively 2^9) whose relevant property is checked by a finite computation, plus a proof that nothing smaller works (18.46) or a general construction (13.19). Two (21.32 and 2.78) are proved by short but non-trivial arguments—a Baer-sum manipulation of central extensions, and a product formula together with an explicit family of p -groups. One (16.60) is a reduction of the question to a classical theorem. In all cases the proofs are complete and elementary, and the computations are reproducible from the repository. The agent was not able to determine whether any of these answers already exist in the literature; the problems were all listed as unsolved in the 21st edition, but 2.78 is sixty years old and 16.60 twenty, and it would not be surprising if parts were known.

The partial results are of a different kind. For 16.14 and 20.37 the agent proved theorems (a structure theorem for minimal counterexamples; a coset-union theorem obtained from the classical perfect-matching property of vertex-transitive graphs) that reduce the problems to finite checks in a range, and then carried out the checks. For 19.56 the reduction to Frattini extensions of minimal simple groups is a clean statement, but the remaining step could not be done. For 20.21 the agent proved two impossibility lemmas and searched well beyond the range of a direct search (its own bound: an example, if it exists, has order at least 6144). In each case the report states precisely what is proved and what is computed.

7.2 Reliability

Three mechanisms supported reliability. (1) *Independent verification of computational claims*: every counterexample or construction was recomputed by a second method, and every 20.37 factorisation was rechecked by brute-force multiplication in GAP. This caught a real bug (the coset-side error) and gave confidence in the others. (2) *Self-review*: on the second day the agent re-read every report line by line, checked each step of the proofs (recorded in the transcript), and corrected several imprecise statements (an unjustified “more generally” clause in the 21.26 note, a vague “chain” claim in the 20.21 note, an overstated list of completely settled groups

in 20.37). (3) *Honest bookkeeping*: the verification table records ranges actually completed, and the notes list literature dependencies that could not be checked.

The obvious weakness is the literature. The agent could search the web but could not read paywalled papers, and did not attempt to find prior work on most of the problems it attacked. A human reviewer should therefore first check whether the “solved” problems are already solved; the mathematics itself should be checkable in a few hours per problem from the appendices.

A second weakness is that all mathematical claims were checked only by the same system that produced them. The proofs in Appendix A are written so that a reader can verify each step; nothing depends on the agent’s authority.

7.3 What worked and what did not

The combination of a computer algebra system with an agent that can write and run code quickly is well suited to this kind of problem list: many notebook problems ask whether a phenomenon occurs, and the SmallGroups library answers such questions up to order 2000 in minutes. The agent’s most productive pattern was “compute first, then prove”: the sharpness of Klyachko’s bound, the class-2 construction for Gorchakov’s problem, and the exclusion strategy for Berkovich’s problem all came from looking at data. Delegating the triage to sub-agents was efficient (about 2300 problems read in an hour) but lossy (one report exceeded the output limit and part of it was lost).

What did not work: attempts at general proofs where the problem is genuinely hard (the general case of 16.14, the Frattini lifting in 19.56, the case $I \cdot I' = K$ of 20.21); the agent recognised this fairly quickly each time and moved to partial results. Two sweeps were started on a vacuous or wrong formulation and had to be redone. Long searches sometimes ran without producing progress output, so their coverage could not be reported precisely, and one (20.21, rank ≥ 5 at order 512) was abandoned as infeasible only after seven hours.

7.4 Recommendations

For anyone repeating such an experiment:

- Provide literature access (a searchable database of MathSciNet/zbMATH reviews or the notebook’s archive of solved problems); the largest uncertainty about the value of the results is their novelty.
- Insist, as here, on independent verification of every computational claim; it is cheap and it works.
- Make the agent write reports as it goes; the reports were the basis for the final self-review and for this paper, and the git history is what allowed the timestamp error to be corrected.
- Budget the final hours explicitly (the agent planned a consolidation phase but mis-scheduled it by misreading a clock); a harness-provided deadline notification would have avoided this.
- Prefer many small, well-instrumented sweeps with progress output over a few large ones.

7.5 Outlook

Several of the partial results suggest natural continuations: for 20.37, the cases $\text{PSL}(3, 4)$ and $\text{Sz}(8)$ with factor sizes 70 and 140 are small enough for an exhaustive (rather than randomised) search and would be the first genuine test of the conjecture beyond subgroup-based constructions; for 16.14, the case $r = 2$ could be closed with the classification of 2-groups with three involutions, and the structure theorem may admit a purely theoretical continuation through the 2-covering group; for 19.56, a uniform treatment of Frattini extensions of minimal simple groups seems within reach of a specialist; for 2.78, the remaining question is whether every $k \geq 9$ is the

number of $\text{IE}\bar{n}$ -systems of some simple group, which depends on the arithmetic of the subgroup orders of $\text{PSL}(2, q)$.

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A Complete proofs for the solved problems

The proofs below are transcriptions of the reports in the repository (`reports/*.md`); they are self-contained.

A.1 Problem 21.32

Throughout, H is a finite group with $H' = G$; we identify G with the subgroup $H' \leq H$ and put $K := C_H(G)$, $Z := Z(G)$.

Lemma A.1. (1) H/G is abelian. (2) $K \trianglelefteq H$ and H/K embeds in $\text{Aut}(G)$ via conjugation; its image $Q := H/K$ contains $\text{Inn}(G) = GK/K$. (3) $K \cap G = Z$. (4) $[H, K] \leq Z$; in particular $K' \leq Z$, K has class ≤ 2 , K/Z is central in H/Z , and K centralises Z .

Proof. (1) $G = H'$. (2) The kernel of the conjugation action $H \rightarrow \text{Aut}(G)$ is $C_H(G) = K$. (3) $K \cap G = C_G(G) = Z$. (4) $[H, K] \leq H' \cap K = G \cap K = Z$ because $K \trianglelefteq H$; the rest follows. $\square$

Theorem A.2 (necessary condition). *If $G \cong H'$ for a finite group H , then there is a subgroup Q with $\text{Inn}(G) \leq Q \leq \text{Aut}(G)$ and $Q' = \text{Inn}(G)$. If $Z(G) = 1$ the converse holds.*

Proof. Put $\tilde{H} = H/Z$, $\tilde{K} = K/Z$, $\tilde{G} = G/Z$. By Lemma A.1(4) $\tilde{K}$ is central in $\tilde{H}$, and $\tilde{H}/\tilde{K} \cong H/K = Q$. Since $H' = G$ we get $\tilde{H}' = \tilde{G}$, and $\tilde{G} \cap \tilde{K} = (G \cap K)/Z = 1$. The projection $\pi : \tilde{H} \rightarrow Q$ maps $\tilde{H}'$ onto Q' with kernel $\tilde{H}' \cap \tilde{K} = 1$, so π restricts to an isomorphism $\tilde{G} = \tilde{H}' \rightarrow Q'$; and $\pi(\tilde{G})$ is the image of G in H/K , i.e. $\text{Inn}(G)$. Hence $Q' = \text{Inn}(G)$. If $Z(G) = 1$ then $K \cap G = 1$ and $[H, K] = 1$, so $K \leq Z(H)$ and $H' \cap K = 1$; hence $H' \cong H'K/K = (H/K)' = Q' = \text{Inn}(G) \cong G$, and conversely, if $Q \leq \text{Aut}(G)$ satisfies $\text{Inn}(G) \leq Q$ and $Q' = \text{Inn}(G) \cong G$, then $H := Q$ works. $\square$

Theorem A.3 (witness bound). *Let $e = 2 \exp(Z)$ and $m = \lceil \log_2 |G| \rceil$. If $G \cong H'$ for some finite H , then $G \cong (H^*)'$ for a finite group H^* with $|H^*| \leq |\text{Aut}(G)| \cdot |Z|^2 \cdot e^{2m}$.*

Proof. Step 1 (bounded rank). $G = H'$ is generated by commutators $[a, b]$ ($a, b \in H$), and every generating set of a group of order $|G|$ contains a generating subset of size $\leq \log_2 |G|$ (a chain of proper subgroups has length $\leq \log_2 |G|$). Choose $a_1, b_1, \dots, a_m, b_m \in H$ with $G = \langle [a_i, b_i] \rangle$ and let $H_0 = \langle G, a_1, b_1, \dots, a_m, b_m \rangle$. Then $G \leq H'_0 \leq H' = G$, so $H'_0 = G$, and H_0/G is abelian, generated by $2m$ elements. Replacing H by H_0 we assume from now on that H/G is abelian and generated by $2m$ elements. Then $K/Z \cong KG/G \leq H/G$ is abelian and generated by $\leq 2m$ elements (subgroups of a d -generated finite abelian group are d -generated).

Step 2 (a large central subgroup). For $k, k' \in K$ we have $[k, k'] \in K' \leq Z$ and K has class ≤ 2 , so $(kk')^e = k^e k'^e [k', k]^{e(e-1)/2}$; since e is even, $e(e-1)/2 = (e/2)(e-1)$ is a multiple of $\exp(Z)$ and the last factor is trivial. Hence $k \mapsto k^e$ is an endomorphism of K and $C := K^e = \{k^e : k \in K\}$ is a characteristic subgroup of K , so $C \trianglelefteq H$. For $h \in H$ and $k \in K$, $[h, k] \in Z$ is centralised by k , so $[h, k^e] = [h, k]^e = 1$ (using $[h, kk'] = [h, k'][h, k]^{k'}$ repeatedly). Thus $C \leq Z(H)$. Furthermore K/C has exponent dividing e , and K/CZ is an abelian group generated by $\leq 2m$ elements of exponent dividing e , so $|K : CZ| \leq e^{2m}$ and

$$|H/C| = |H : K| \cdot |K : CZ| \cdot |CZ : C| \leq |\text{Aut}(G)| \cdot e^{2m} \cdot |Z|. \quad (1)$$

Step 3 (replacing C by $C \cap G$). Put $\bar{H} := H/C$ and $C_0 := C \cap G = C \cap Z$. H is a central extension $1 \rightarrow C \rightarrow H \rightarrow \bar{H} \rightarrow 1$ with class $\xi \in H^2(\bar{H}, C)$ (trivial action). We use two standard facts about a central extension $1 \rightarrow M \rightarrow E \rightarrow Q \rightarrow 1$ with class η :

(F1) (Hopf) the five-term exact sequence $H_2(E) \rightarrow H_2(Q) \rightarrow M \rightarrow E^{ab} \rightarrow Q^{ab} \rightarrow 0$ shows that the image of the Hopf map $h_\eta : H_2(Q) \rightarrow M$ equals $M \cap E'$;

(F2) (universal coefficients) $0 \rightarrow \text{Ext}(Q^{ab}, M) \rightarrow H^2(Q, M) \rightarrow \text{Hom}(H_2(Q), M) \rightarrow 0$ is exact, the second map sending η to h_η ; the sequence is natural in M .

By (F1), h_ξ has image $C \cap H' = C_0$. Let $\iota : C_0 \hookrightarrow C$. By (F2) with coefficient group C_0 there is $\xi^* \in H^2(\bar{H}, C_0)$ with $h_{\xi^*} = h_\xi$ (as a map into C_0). Let H^* be a central extension of $\bar{H}$ by C_0 with class ξ^* ; then $|H^*| = |\bar{H}| \cdot |C_0|$. By naturality $h_{\iota_* \xi^*} = \iota \circ h_{\xi^*} = h_\xi$, so $\xi - \iota_* \xi^*$ has trivial Hopf map, i.e. lies in $\text{Ext}(\bar{H}^{ab}, C)$; let A be a central extension of $\bar{H}$ by C with this class. By (F1), $A' \cap C = 1$, so $A' \rightarrow \bar{H}'$ is an isomorphism.

The extension with class $\iota_* \xi^*$ is the pushout $E_1 := (H^* \times C)/\{(c, c^{-1}) : c \in C_0\}$; the map $H^* \rightarrow E_1$ is injective, C is central in E_1 , $E_1 = H^*C$, and hence $E'_1 = (H^*)'$.

Now $\xi = \iota_* \xi^* + (\xi - \iota_* \xi^*)$ is a Baer sum: $H \cong F/\Delta$ where $F := E_1 \times_{\bar{H}} A$ is the fibre product over $\bar{H}$ (a central extension of $\bar{H}$ by $C \times C$) and $\Delta = \{(c, c^{-1}) : c \in C\}$.

Claim: $F' = E'_1 \times_{\bar{H}'} A' \cong E'_1$. Indeed $F' \subseteq E'_1 \times_{\bar{H}'} A' := \{(x, y) \in E'_1 \times A' : x, y \text{ have the same image in } \bar{H}'\}$. By naturality of the Hopf map under the two projections, the Hopf map of F is $(h_{\iota_* \xi^*}, h_{\xi - \iota_* \xi^*}) = (h_\xi, 0)$, with image $C_0 \times 1$; by (F1), $F' \cap (C \times C) = C_0 \times 1$. As F' maps onto $\bar{H}'$, $|F'| = |\bar{H}'| \cdot |C_0|$. On the other hand $|E'_1 \times_{\bar{H}'} A'| = \sum_{z \in \bar{H}'} |E'_1 \cap \pi^{-1}(z)| \cdot |A' \cap \pi^{-1}(z)| = |\bar{H}'| \cdot |E'_1 \cap C| \cdot |A' \cap C| = |\bar{H}'| \cdot |C_0| \cdot 1$. Hence $F' = E'_1 \times_{\bar{H}'} A'$, and the projection to

the first coordinate is an isomorphism onto E'_1 (injective because $A' \cap C = 1$, surjective because $A' \rightarrow \bar{H}'$ is onto).

Finally $F' \cap \Delta = \{(c, c^{-1}) : c \in C_0, c^{-1} = 1\} = 1$, so $H' = (F/\Delta)' = F'\Delta/\Delta \cong F' \cong E'_1 = (H^*)'$. Therefore $(H^*)' \cong H' = G$ and, by (1), $|H^*| = |\bar{H}| \cdot |C_0| \leq |\text{Aut}(G)| \cdot e^{2m} \cdot |Z| \cdot |Z|$. $\square$

Remark A.4. The witness has order $|G|^{O(\log |G|)}$ (as $|\text{Aut } G| \leq |G|^{\log_2 |G|}$), and “ $(H^*)' \cong G$ ” is verifiable in time quasi-polynomial in $|G|$ (group isomorphism for groups of order N is decidable in time $N^{O(\log N)}$); so the decision problem is in nondeterministic quasi-polynomial time. For centreless G Theorem A.2 gives a much better algorithm inside $\text{Aut}(G)$. Examples: $\text{Aut}(S_3) = S_3$ has derived subgroup $C_3 \neq \text{Inn}(S_3)$, and no subgroup of $\text{Aut}(D_8) \cong D_8$ containing $\text{Inn}(D_8) \cong C_2^2$ has derived subgroup C_2^2 ; so S_3 and D_8 are not derived subgroups of finite groups (confirmed by brute force in `work/p21_32/`).

A.2 Problem 16.60

Theorem A.5. *For every finite group G and every $\alpha \in \text{Aut}(G)$, $|S_\alpha| \leq T(G)$.*

Proof. Step 1. Put $C := C_G(\alpha^2) = \{g : \alpha^2(g) = g\}$. If $g \in S_\alpha$ then $\alpha^2(g) = \alpha(g^{-1}) = \alpha(g)^{-1} = g$, so $S_\alpha \subseteq C$. Since α commutes with α^2 , $\alpha(C) = C$, and $\sigma := \alpha|_C \in \text{Aut}(C)$ satisfies $\sigma^2 = \text{id}$. Clearly $S_\alpha = \{g \in C : \sigma(g) = g^{-1}\} =: S_\sigma(C)$. It suffices to prove (A) $|S_\sigma(C)| \leq T(C)$ for every finite group C and involutory $\sigma \in \text{Aut}(C)$, and (B) $T(C) \leq T(G)$ for subgroups $C \leq G$.

Step 2 (B). For $\theta \in \text{Irr}(C)$ the induced character θ^G is a non-zero character, so $m(\theta) := \sum_{\chi \in \text{Irr}(G)} \langle \chi, \theta^G \rangle \geq 1$. By Frobenius reciprocity, $T(G) = \sum_{\chi} \chi(1) = \sum_{\chi} \sum_{\theta} \langle \chi_C, \theta \rangle \theta(1) = \sum_{\theta} \theta(1)m(\theta) \geq \sum_{\theta} \theta(1) = T(C)$.

Step 3 (A). For $\theta \in \text{Irr}(C)$ put $\varepsilon_\sigma(\theta) := \frac{1}{|C|} \sum_{g \in C} \theta(g\sigma(g))$. (i) By the second orthogonality relation $\sum_{\theta} \theta(1)\theta(x) = |C|[x = 1]$, $\sum_{\theta} \varepsilon_\sigma(\theta)\theta(1) = \#\{g \in C : g\sigma(g) = 1\} = |S_\sigma(C)|$. (ii) $\varepsilon_\sigma(\theta) \in \{1, 0, -1\}$: let $\rho : C \rightarrow \text{GL}(V)$ afford θ and $\rho^\sigma := \rho \circ \sigma$. For linear maps A, B on V , $\text{tr}(AB) = \text{tr}(\tau \circ (A \otimes B))$, where τ is the flip on $V \otimes V$. Hence $\varepsilon_\sigma(\theta) = \text{tr}(\tau \circ P)$ where $P = \frac{1}{|C|} \sum_g (\rho \otimes \rho^\sigma)(g)$ is the projection onto the invariants W of $\rho \otimes \rho^\sigma$. Since $\sigma^2 = 1$, $\tau \circ (\rho \otimes \rho^\sigma)(g) = (\rho \otimes \rho^\sigma)(\sigma(g)) \circ \tau$, so τ maps W into itself and $\varepsilon_\sigma(\theta) = \text{tr}(\tau|_W)$, an integer of absolute value $\leq \dim W = \dim \text{Hom}_C(V^*, V^\sigma) = \langle \bar{\theta}, \theta^\sigma \rangle \leq 1$. (iii) From (i) and (ii), $|S_\sigma(C)| = \sum_{\theta} \varepsilon_\sigma(\theta)\theta(1) \leq \sum_{\theta} \theta(1) = T(C)$. $\square$

Step 3 is the twisted Frobenius–Schur theorem of Kawanaka and Matsuyama [2]; Step 1 is the only new ingredient. A numerical check for all groups of order ≤ 200 and all automorphisms is in `work/p16_60/`.

A.3 Problem 2.78

For a finite group G let $\text{Ord}(G)$ be the set of orders of subgroups of G , $f(G) = \#\{d : G \text{ has a non-normal subgroup of order } d\}$ and $n(G) = |\text{Ord}(G)| - f(G)$ (the number of orders d all of whose subgroups are normal).

Lemma A.6 (sections). *If $S = H/N$ with $N \trianglelefteq H \leq G$, then $f(G) \geq f(S)$. For non-abelian simple S , $f(S) = |\text{Ord}(S)| - 2$.*

Proof. If K/N ($N \leq K \leq H$) is non-normal in S then K is not normal in G (else $K \trianglelefteq H$ and $K/N \trianglelefteq S$), and distinct $|K/N|$ give distinct $|K| = |N||K/N|$. For simple S every proper non-trivial subgroup is non-normal. $\square$

Lemma A.7 (coprime direct products). *If $\gcd(|G|, |H|) = 1$ then $f(G \times H) = |\text{Ord}(G)||\text{Ord}(H)| - n(G)n(H)$.*

Proof. Every subgroup of $G \times H$ is $A \times B$ with $A \leq G$, $B \leq H$, normal iff $A \trianglelefteq G$ and $B \trianglelefteq H$; distinct pairs $(|A|, |B|)$ give distinct products. So the order ab carries a non-normal subgroup iff some subgroup of order a is non-normal in G or some subgroup of order b is non-normal in H . $\square$

In particular, for $G = A_5$ ($\text{Ord}(A_5) = \{1, 2, 3, 4, 5, 6, 10, 12, 60\}$, $n(A_5) = 2$): $f(A_5 \times H) = 9|\text{Ord}(H)| - 2n(H)$ whenever $\gcd(|H|, 30) = 1$.

Lemma A.8 (a family of p -groups). *Let p be an odd prime, $n \geq 3$, $r \geq 0$, $M = M_{p^n} = \langle x, y \mid x^{p^{n-1}} = y^p = 1, x^y = x^{1+p^{n-2}} \rangle$ and $P = M \times C_{p^r}$. Then $|\text{Ord}(P)| = n+r+1$ and $f(P) = r+1$: the non-normal subgroups of P have exactly the orders $p, p^2, \dots, p^{r+1}$.*

Proof. P is a p -group of order p^{n+r} , so all orders p^i ($0 \leq i \leq n+r$) occur. Put $z = x^{p^{n-2}}$; then $M' = \langle z \rangle$ has order p , $Z(M) = \langle x^p \rangle$, and $C = C_{p^r}$ is central in P . For $U \leq P$ with projection $V = \text{pr}_M(U)$ we have $[U, P] = [V, M] \times 1$, which is 1 if $V \leq Z(M)$ and $\langle z \rangle \times 1$ otherwise; so $U \trianglelefteq P$ iff $V \leq Z(M)$ or $(z, 1) \in U$. (i) The subgroups $U = \langle (y, 1) \rangle \times \langle (1, c) \rangle$ with $c \in C$ of order p^j ($0 \leq j \leq r$) satisfy $V = \langle y \rangle \not\leq Z(M)$ and contain no element $(z, 1)$ (their elements are (y^i, c^k) and $z \notin \langle y \rangle$), so they are non-normal of order p^{j+1} . (ii) Let U be non-normal, so $V \not\leq Z(M)$ and $(z, 1) \notin U$. V is abelian (otherwise U would contain a commutator $([m, m'], 1) = (z^j, 1) \neq 1$), so $V \leq C_M(v)$ for some $v \in V \setminus Z(M)$, and $C_M(v) = \langle v, Z(M) \rangle$ has order p^{n-1} (the centraliser of a non-central element has index p since $M/Z(M) \cong C_p^2$). If $v = x^i y^j$ with $p \nmid i$ then $v^p = x^{ip}$ (p odd, class 2, $|M'| = p$), so v has order p^{n-1} and $C_M(v) = \langle v \rangle$ is cyclic with unique subgroup of order p equal to $\langle z \rangle$; if $p \mid i$ then $v = x^{pi'} y^j$ with $p \nmid j$, and replacing v by a power generating the same subgroup we may assume $v = yx^{pi'}$; then $C_M(v) = \langle y \rangle \times \langle x^p \rangle$ with $z \in \langle x^p \rangle$. In both cases a subgroup $V_0 \leq C_M(v)$ with $z \notin V_0$ meets the cyclic factor $\langle x^p \rangle$ (resp. $\langle v \rangle$) trivially, hence $|V_0| \leq p$. Now $U \cap (M \times 1) = V_0 \times 1$ with $z \notin V_0$, so $|V_0| \leq p$ and $|U| = |V_0| \cdot |\text{pr}_C(U)| \leq p^{r+1}$. $\square$

(The lemma was also checked by computer for $p = 7, 11$, $n \leq 4$, $r \leq 2$.)

Theorem A.9. *Every $k \geq 86$ is $f(G)$ for a non-soluble non-simple group G . Hence the soluble numbers and the absolutely simple numbers form finite sets (contained in $\{0, \dots, 85\}$).*

Proof. The numbers $7n + 9$ for $n = 3, \dots, 11$ are 30, 37, 44, 51, 58, 65, 72, 79, 86, which represent all residue classes modulo 9 and are ≤ 86 . Hence every $k \geq 86$ can be written as $k = 7n + 9r + 9$ with $3 \leq n \leq 11$ and $r \geq 0$. Take a prime $p \geq 7$ and $P = M_{p^n} \times C_{p^r}$. By Lemmas A.7 and A.8, $f(A_5 \times P) = 9(n+r+1) - 2n = 7n + 9r + 9 = k$, and $A_5 \times P$ is non-soluble and not simple. $\square$

Theorem A.10. (a) $f(G) \geq 7$ for every non-soluble G ; (b) $f(G) = 7$ iff $G \cong A_5$; (c) no non-soluble group has $f(G) = 8$; (d) $f(S) \geq 9$ for every non-abelian simple $S \neq A_5$.

Proof. (d) Every non-abelian simple S has a minimal non-soluble subgroup U , and $U/\Phi(U)$ is a minimal simple group T (Thompson [7]); by Lemma A.6, $f(S) \geq f(T)$. We check $f(T) \geq 9$, i.e. $|\text{Ord}(T)| \geq 11$, for every minimal simple $T \neq A_5$: for $T = \text{PSL}(2, 2^p)$ ($p \geq 3$ prime) $\text{Ord}(T)$ contains the $p+1$ powers of 2 up to 2^p , the divisors of $2^p \pm 1$ and their doubles (cyclic and dihedral subgroups), the Borel order $2^p(2^p - 1)$ and $|T|$; for $p = 3$ this gives $|\text{Ord}| = 12$, and for $p \geq 5$ at least $6 + 6$. For $T = \text{PSL}(2, 3^p)$ (p odd prime) similarly $|\text{Ord}(\text{PSL}(2, 27))| = 14$ and more for larger p . For $T = \text{PSL}(2, p)$, $p \geq 7$ prime, $\text{Ord}(T)$ contains $1, 2, 3, 4, 6, 12, p, (p-1)/2, (p+1)/2, p-1, p+1, p(p-1)/2, |T|$ (Klein four groups, S_3 , A_4 , cyclic and dihedral subgroups, the Borel subgroup); these 13 numbers are distinct unless $p \in \{7, 11, 13, 23\}$; in those cases $\text{Ord}(\text{PSL}(2, 7)) = \{1, 2, 3, 4, 6, 7, 8, 12, 21, 24, 168\}$ (using $S_4 \leq \text{PSL}(2, 7)$), $\text{Ord}(\text{PSL}(2, 11)) \supseteq \{1, 2, 3, 4, 5, 6, 10, 11, 12, 55, 60, 660\}$ ($A_5 \leq \text{PSL}(2, 11)$), $\text{Ord}(\text{PSL}(2, 13)) \supseteq \{1, 2, 3, 4, 6, 7, 12, 13, 14, 26, 39, 78, 1092\}$ and $\text{Ord}(\text{PSL}(2, 23)) \supseteq \{1, 2, 3, 4, 6, 8, 11, 12, 22, 23, 24, 253, 6072\}$, so $|\text{Ord}| \geq 11$ throughout. For $T = \text{Sz}(2^p)$ there are $2p+1$ two-power orders, three odd torus orders and $|T|$, so $|\text{Ord}| \geq 2p+5 \geq 11$. For $T =$

$\text{PSL}(3, 3)$, $|\text{Ord}| = 23$. So $f(S) \geq 9$ unless the minimal simple section is A_5 . If $U/\Phi(U) \cong A_5$ with $\Phi(U) \neq 1$, then besides the 7 orders $|\Phi(U)|d$ ($d \in D := \{2, 3, 4, 5, 6, 10, 12\}$) the subgroups $\Phi(U)$ and U give two further non-normal orders, so $f(S) \geq 9$. If $A_5 \cong U \leq S \neq A_5$, then $\text{Ord}(S) \supseteq \text{Ord}(A_5) \cup \{60\}$ gives $f(S) \geq 8$, and a Sylow 2-subgroup of order ≥ 8 gives a ninth order unless $|S|_2 = 4$; then $S \cong \text{PSL}(2, q)$ with $q \equiv \pm 3 \pmod{8}$ (Gorenstein–Walter), $q \geq 11$, and $q \in \text{Ord}(S)$ is a new order.

(a) follows from Lemma A.6, (d) and $f(A_5) = 7$.

(b), (c). Let G be non-soluble with $f(G) \in \{7, 8\}$. By (d) and Lemma A.6 every non-abelian composition factor of G is A_5 ; fix $H \leq G$, $N \trianglelefteq H$ with $H/N \cong A_5$. The 7 orders $|N|d$ ($d \in D$) are orders of non-normal subgroups of G ; at most one further order e is available.

$N = 1$. For $p \in \{2, 3, 5\}$ a Sylow p -subgroup Q of H is non-normal in G (its image in H/N is a non-normal Sylow subgroup of A_5), and $|Q| = |A_5|_p |N|_p$. If $|Q| = |N|d$ with $d \in D$ then $|N|_p d = |A_5|_p$, forcing $d = |A_5|_p$ and N a p -group. If $N \neq 1$, N is a p -group for at most one of the three primes, so at least two of the numbers $4|N|_2, 3|N|_3, 5|N|_5$ would have to equal the single extra order e ; but no two of them are equal. Hence $N = 1$ and $A_5 \cong H \leq G$.

Minimal normal subgroups. Let M be a minimal normal subgroup of G . If $M \cap H = 1$ then for each non-normal $K \leq H$ the subgroup KM is non-normal in G ($KM \trianglelefteq G$ would give $KM \cap H = K(M \cap H) = K \trianglelefteq H$), of order $|K||M|$ with $|M| \geq 2$; the two orders $10|M| \geq 20$ and $12|M| \geq 24$ lie outside D and are distinct, so two extra orders would be needed. Hence $M \geq H$, so $M \cong A_5^k$ (all non-abelian composition factors are A_5); $k \geq 2$ is impossible since $A_5 \times A_5$ has non-normal subgroups of the nine orders $2, 3, 4, 5, 6, 9, 10, 12, 25$. Thus $M = A_5$ is the unique minimal normal subgroup, $C_G(M) = 1$ and G embeds in $\text{Aut}(A_5) = S_5$. As $f(S_5) = 10$, $G = A_5$ and $f(G) = 7$. So no non-soluble group has $f = 8$, and $f = 7$ forces $G \cong A_5$. $\square$

Remark A.11. The values $f(G)$ attained by non-soluble groups of order ≤ 1000 are $7, 9, 10, 12, \dots, 18, 20, \dots, 23$; small simple groups give $f(\text{PSL}(2, 7)) = 9$, $f(\text{PSL}(2, 8)) = 9$, $f(\text{PSL}(2, 11)) = 10$, $f(\text{PSL}(2, 13)) = 11$, $\dots$, $f(M_{11}) = 24$, and $f(\text{PSL}(2, q))$ for $q \leq 169$ takes the values $7, 9\text{--}23, 25, 26, 28, 32, 35, 42$; together with $f(M_{11}) = 24$ every integer in $\{7, 9, \dots, 26\} \setminus \{8\}$ is a simple number. Whether composite non-soluble numbers exist (i.e. whether some $k \geq 9$ is avoided by all simple groups) remains open.

A.4 Problem 18.46

Theorem A.12. *Let H be a finite group containing a subgroup $D \cong D_8$ such that every element of D is a square in H . Then $|H| \geq 128$; the wreath product $H = D_8 \wr C_2 = (D_8 \times D_8) \rtimes \langle \sigma \rangle$ attains the bound, the diagonal subgroup $\{(g, g)\} \cong D_8$ consisting of squares since $((g, 1)\sigma)^2 = (g, 1)(1, g) = (g, g)$.*

Proof. If $|H| < 128$ and $D_8 \leq H$ then $8 \mid |H|$, so $|H| \in \{8, 16, \dots, 120\}$; all groups of these orders are in the SmallGroups library. Two independent GAP computations (`work/p18_46/check.g` using `IsomorphicSubgroups(H, D8)`, and `verify_d8.g` using the full subgroup lattice and `IdGroup`) show that no group of order $8, 16, \dots, 120$ has a subgroup $S \cong D_8$ with $S \subseteq \{h^2 : h \in H\}$. Exactly 14 of the 2328 groups of order 128 (SmallGroup ids 134, 136, 138, 139, 140, 141, 144, 146, 928–933) do, among them $D_8 \wr C_2 = \text{SmallGroup}(128, 928)$. $\square$

The same computation gives the minimal $|H|$ for the sixteen groups of order ≤ 12 listed in the report (the remaining groups of order ≤ 12 have odd order, where $H = G$ works since every element is a square, or are C_{10} , where $H = C_{20}$ works); D_8 is the only one of them needing $|H| > |G|^2$ (e.g. C_2^2 : 16, S_3 : 36, Q_8 : 48, C_3^3 : 64, A_4 : 24, D_{12} : 144). Hence Klyachko’s estimate $|H| \leq 2|G|^2$ cannot be improved to $c|G|^2$ with $c < 2$.

A.5 Problem 13.19

Reformulation. If $Q \leq G_1 \times \cdots \times G_n$ and $N_i = Q \cap \ker \pi_i$, then $\bigcap N_i = 1$, and “ H projects onto G_i ” means $HN_i = Q$. Conversely, a group Q with normal subgroups $N_1, \dots, N_n$ with $\bigcap N_i = 1$ and a normal subgroup H with $HN_i = Q$ for all i gives such a configuration with $G_i = Q/N_i$. One checks $\gamma_n(Q) \leq H \cdot \bigcap_i N_i = H$ (a commutator of weight n has, for each i , a factor that can be taken in N_i modulo H), so Q/H has class $\leq n - 1$; in particular $n \geq 3$ is needed for a non-regular 2-group Q/H .

The explicit example ($n = 3, p = 2$). Let $V = \mathbb{F}_2^2$ with the alternating form $\gamma(v, w) = v_1w_2 + v_2w_1$ and the bilinear map $c_1(v, w) = v_1w_2$, and let $A = \mathbb{F}_2^3$ with basis u_{12}, u_{13}, u_{23} ; put $w_{11} = u_{12}, w_{22} = u_{23}, w_{33} = u_{13}, w_{ij} = w_{ji} = u_{ij}$ ($i < j$). On $Q = V^3 \times A$ define $(x, a)(y, b) = (x + y, a + b + c(x, y))$ with $c(x, y) = \sum_i c_1(x_i, y_i)w_{ii} + \sum_{i < j} \gamma(x_i, y_j)w_{ij}$. Since c is bilinear this is a group of order 2^9 (a central extension of V^3 by A), generated by involutions e_i, f_i (the standard basis of the i -th copy of V) and central involutions u_{12}, u_{13}, u_{23} , with $[e_i, f_i] = w_{ii}, [e_i, f_j] = [f_i, e_j] = u_{ij}$ ($i \neq j$), all other commutators trivial. Put

$$N_i = \{(x, a) : x_j = 0 \ (j \neq i), \ a \in \langle u_{ij}, u_{ik} \rangle\} \quad (\{i, j, k\} = \{1, 2, 3\}),$$

$$\hat{H} = \{(x, a) : x_1 + x_2 + x_3 = 0\}, \quad A_0 = \langle u_{12} + u_{13}, u_{13} + u_{23} \rangle.$$

Then: (1) N_i is a normal subgroup of order 16, $N_i \cap N_j = \langle u_{ij} \rangle$ and $N_1 \cap N_2 \cap N_3 = 1$; (2) $\hat{H}$ is a subgroup of order 2^7 containing A , and $\hat{H}/A_0$ is elementary abelian and central in Q/A_0 (modulo A_0 all w_{ij} become equal, so commutators and squares of elements with $\sum x_i = 0$ vanish); (3) choosing a complement H/A_0 of A/A_0 in $\hat{H}/A_0$ gives a normal subgroup H of order 2^6 with $H \cap A = A_0$; (4) $HN_i = Q$ for all i (as $H \cap N_i = A_0 \cap N_i$ has order 2); (5) Q/H is a non-abelian group of order 8, namely D_8 (the commutator form induced by $\gamma(\sum x_i, \sum y_i)$ is non-degenerate and the squaring map is isotropic). With $G_i = Q/N_i$ (of order 32) this is a configuration as in the problem with $Q/H \cong D_8$, which is not regular. All of (1)–(5) were verified in GAP from the pc presentation (`work/p13_19/construct_pc2.g`) with $H = \langle e_1e_2u_{23}, f_1f_2u_{23}, e_2e_3u_{23}, f_2f_3u_{23}, u_{12}u_{23}, u_{13}u_{23} \rangle$.

The general construction.

Lemma A.13 (vanishing of the top difference). *Let G be nilpotent of class c and w a word in $n > c$ blocks of variables. For a tuple a of elements of G (grouped in n blocks) and $S \subseteq [n]$ let a_S be a with the entries of the blocks outside S replaced by 1. If $w(a_S) = 1$ for every proper $S \subsetneq [n]$, then $w(a) = 1$.*

Proof. Let N be the free nilpotent group of class c on the variables, $\sigma_S : N \rightarrow N$ the endomorphism killing the variables outside S (so $\sigma_S \sigma_T = \sigma_{S \cap T}$), and M the normal closure of $\{\sigma_S(w) : S \subsetneq [n]\}$. Since $\sigma_T(M) \leq M$, the σ_T act on $\bar{N} = N/M$, and $\sigma_S(\bar{w}) = 1$ for all proper S . Suppose $\bar{w} \in \gamma_k(\bar{N})$ with $k \leq c$ and let $A = \gamma_k(\bar{N})/\gamma_{k+1}(\bar{N})$, spanned by weight- k commutators in the generators; for $T \subseteq [n]$ let A_T be the span of those involving exactly the blocks in T . Then σ_S acts on A_T as the identity if $T \subseteq S$ and as 0 otherwise, and $\bar{w}\gamma_{k+1} = \sum_T a_T$ with $a_T \in A_T$, where only $|T| \leq k < n$ occurs. Hence $0 = \sigma_S(\bar{w}\gamma_{k+1}) = \sum_{T \subseteq S} a_T$ for every proper S , and since $\sum_{T \subseteq S \subsetneq [n]} (-1)^{n-|S|+1} = 1$ for every proper T , we get $0 = \sum_{S \subsetneq [n]} (-1)^{n-|S|+1} \sigma_S(\bar{w}\gamma_{k+1}) = \sum_T a_T = \bar{w}\gamma_{k+1}$. So $\bar{w} \in \gamma_{k+1}(\bar{N})$, and inductively $\bar{w} \in \gamma_{c+1}(\bar{N}) = 1$. The homomorphism $N \rightarrow G$ sending the variables to a maps M to 1 by hypothesis, hence factors through $\bar{N}$, and $w(a)$ is the image of $\bar{w} = 1$. $\square$

Theorem A.14. *Let P be a finite p -group of class c and $n = c + 1$. Let $D = \prod_{\emptyset \neq S \subseteq [n]} P^{(S)}$; for a generator g of P and $i \in [n]$ let $g^{[i]} \in D$ have S -component g if $i \in S$ and 1 otherwise; let Q be the subgroup generated by all $g^{[i]}$, $H = Q \cap \ker \pi_{[n]}$ and $N_i = Q \cap \bigcap_{S \not\ni i} \ker \pi_S$. Then $Q/H \cong P$,*

$HN_i = Q$ for all i , and $N_1 \cap \dots \cap N_n = 1$; so Q , H and $G_i = Q/N_i$ form a configuration as in Problem 13.19 with $Q/H \cong P$.

Proof. $\pi_{[n]}(g^{[i]}) = g$, so $\pi_{[n]}(Q) = P$ and $Q/H \cong P$. N_i is normal (an intersection of Q with kernels), $g^{[i]} \in N_i$, so $\pi_{[n]}(N_i) = P$ and $HN_i = Q$. If $q \in \bigcap N_i = Q \cap P^{([n])}$ then q is a word w in the tagged generators with $\pi_S(q) = w(a_S) = 1$ for all proper S (a the tuple of generators grouped in n blocks), so $\pi_{[n]}(q) = w(a) = 1$ by Lemma A.13, i.e. $q = 1$. $\square$

Taking $P = C_p \wr C_p$ (class p , irregular) gives a negative answer for every prime p . The construction was checked in GAP for $P = D_8$ ($|Q| = 2^{12}$) and $P = C_3 \wr C_3$ ($|Q| = 3^{32}$).

B Theorems and proofs for the partial results

B.1 Problem 16.14

Notation: G is a finite 2-group with $\Omega_1(G) \leq Z(G)$; $E = \Omega_1(G)$ (elementary abelian, central, generated by all involutions), $r = \text{rank } E = \text{rank } Z(G)$, $d = d(G) = \text{rank}(G/G^2) = \dim G/\Phi(G)$.

Theorem B.1 (class ≤ 2). *If G has class ≤ 2 and $G^2 \leq Z(G)$, then $d(G) \leq 2 \text{rank } Z(G)$.*

Proof. Put $Z = Z(G)$ and $r = \text{rank } Z = \dim(Z/Z^2)$. Since G has class ≤ 2 , $(gh)^2 = g^2h^2[h, g]$ with $[h, g] \in G' \leq Z$. Hence $Q : G \rightarrow Z/Z^2$, $g \mapsto g^2Z^2$, satisfies $Q(gh) = Q(g)Q(h)[h, g]Z^2$, $Q(gz) = Q(g)$ for $z \in Z$, $Q(g^2) = 1$. So Q factors through the $\mathbb{F}_2$ -space $V = G/(ZG^2)$ and is a quadratic map $q : V \rightarrow W = Z/Z^2 \cong \mathbb{F}_2^r$ with polar form the commutator pairing; in coordinates $q = (q_1, \dots, q_r)$. If $v \in V \setminus \{0\}$ had $q(v) = 0$, a preimage $g \notin ZG^2$ would satisfy $g^2 = z^2$ with $z \in Z$, so $(gz^{-1})^2 = 1$, $gz^{-1} \in \Omega_1(G) \leq Z$ and $g \in Z$, a contradiction. So $q_1, \dots, q_r$ have no common non-trivial zero, and by the Chevalley–Warning theorem $n := \dim V \leq 2r$. With $W' = \text{span } q(V)$ and $r' = \dim W' \leq r$ the same argument gives $n \leq 2r'$. Finally $\Phi(G) = G^2G' \leq Z$, $Z^2 \leq G^2 \leq \Phi(G)$ and $\Phi(G)Z^2/Z^2 \supseteq W'$, so $d(G) = \dim G/ZG^2 + \dim ZG^2/\Phi(G) = n + \dim Z/\Phi(G) \leq n + (r - r') \leq 2r' + r - r' \leq 2r$. $\square$

If $\Phi(G)$ is central and elementary abelian (p -class ≤ 2), the theorem applies.

Theorem B.2 (minimal counterexample). *Let G be a counterexample of minimal order ($d(G) > 2r(G)$ and $d(K) \leq 2r(K)$ for all smaller K with $\Omega_1(K) \leq Z(K)$), $E = \Omega_1(G)$, $r = r(G)$, $H = G/E$. Then:*

- (i) $d(G) = 2r + 1$, and every maximal subgroup M satisfies $\Omega_1(M) = E$ and $d(M) = 2r = 2r(M)$;
- (ii) $E \leq \Phi(G)$, hence $d(H) = 2r + 1$;
- (iii) $\Phi(M) = \Phi(G)$ for every maximal M , hence $\Phi(\bar{M}) = \Phi(H)$ for every maximal $\bar{M} \leq H$;
- (iv) every $g \in G \setminus E$ has order $2 \cdot |gE|$;
- (v) for every elementary abelian $A \leq H$ the squaring map induces a quadratic map $q_A : A \rightarrow E$ without non-trivial zeros; hence $\text{rank } A \leq 2r$;
- (vi) every subgroup of H with trivial Schur multiplier is abelian;
- (vii) $r \geq 2$, $|\Phi(G)| \geq 2|E|$ and $|G| \geq 2^{3r+2}$.

Proof. Subgroups inherit the hypothesis: for $K \leq G$, $\Omega_1(K) \leq E \cap K \leq Z(K)$ and $r(K) \leq r$. For a maximal subgroup M , $M/\Phi(G)$ is an elementary abelian quotient of M of rank $d - 1$, so $d(M) \geq d(G) - 1$; by minimality $d(M) \leq 2r(M) \leq 2r$, hence $d(G) \leq 2r + 1$, so $d(G) = 2r + 1$, $d(M) = 2r$, $r(M) = r$ and $\Omega_1(M) = E$: (i). (ii) If $t \in E \setminus \Phi(G)$, pick a maximal M with $t \notin M$; then $G = M \times \langle t \rangle$, $r(M) = r - 1$ and $d(M) = 2r = 2r(M) + 2$, contradicting minimality for M . (iii) $\Phi(M) \leq \Phi(G)$ (M normal) and $|M : \Phi(M)| = 2^{2r} = |M : \Phi(G)|$. (iv) Let $|gE| = 2^k$;

then $g^{2^k} \in E$ and $g^{2^{k-1}} \notin E$; if $g^{2^k} = 1$ then $g^{2^{k-1}}$ would be an involution outside E . (v) The preimage $\tilde{A}$ of A has $\tilde{A}^2 \leq E$, $\tilde{A}' \leq E \leq Z(\tilde{A})$, so $q_A(aE) = a^2$ is a well-defined quadratic map with polar form the commutator pairing; $q_A(aE) = 0$ with $aE \neq 0$ would give an involution $a \notin E$. Chevalley–Warning gives $\text{rank } A \leq 2r$. (vi) For $Q \leq H$ with preimage $\tilde{Q}$, the five-term sequence $H_2(\tilde{Q}) \rightarrow H_2(Q) \rightarrow E \rightarrow \tilde{Q}^{ab} \rightarrow Q^{ab} \rightarrow 0$ gives $E \cap \tilde{Q}' = \text{im}(H_2(Q) \rightarrow E) = 0$, so $\tilde{Q}' \cong Q'$; if $Q' \neq 1$ the 2-group $\tilde{Q}'$ contains an involution outside E . (vii) $r = 1$ would make G cyclic or generalised quaternion ($d \leq 2$); $\Phi(G) = E$ would make $\Phi(G)$ central elementary abelian and Theorem B.1 would apply. So $|\Phi(G)| \geq 2^{r+1}$ and $|G| = 2^d |\Phi(G)| \geq 2^{3r+2}$. $\square$

Theorem B.3 (small orders). *There is no counterexample of order $\leq 2^{11}$, and none of order 2^{12} with $r \geq 3$.*

Proof (computation guided by Theorem B.2). By (vii) a minimal counterexample has $|G| = 2^{2r+1} |\Phi(G)|$ with $|\Phi(G)| \geq 2^{r+1}$, $r \geq 2$: for $r = 2$, $|G| \geq 2^8$; for $r = 3$, $|G| \geq 2^{11}$; for $r \geq 4$, $|G| \geq 2^{14}$. Order 2^8 is excluded by the direct sweep of all groups of order 256. For the remaining cases $H = G/E$ is a group of order $|G|/2^r \in \{2^7, 2^8, 2^9\}$ with $d(H) = 2r + 1$, available in the SmallGroups library, and G is a central extension of H by $E = \mathbb{F}_2^r$ with $E \leq \Phi(G)$ in which, by (iv), no involution of H lifts to an involution. Every such G is a quotient of the 2-covering group $P^* = F/R^2[F, R]$ of $H = F/R$. Hence H is excluded as soon as some involution of H lifts to an involution of P^* (filter F3', computed with ANUPQ's PqPCover; it is equivalent to the vanishing on all 2-cocycles of the functional “square of the lift”, which was used in the first version of the computation), or some maximal subgroup $\bar{M}$ of H has $\Phi(\bar{M}) \neq \Phi(H)$ (filter F2, by (iii)), or H has an elementary abelian subgroup of rank $2r + 1$ (filter F1, by (v)). Surviving H would require a search for r cohomology classes satisfying the lifting condition; none survived. The cases: $r = 3$, $|G| = 2^{11}$: the 10 groups of order 256 with $d = 7$ (two fail F1, eight F3); $r = 3$, $|G| = 2^{12}$: the 1139 groups of order 512 with $d = 7$ (32 fail F1, 82 F2, 1025 F3); $r = 2$, $|G| = 2^9$: the 169 groups of order 128 with $d = 5$; $r = 2$, $|G| = 2^{10}$: the 28 653 groups of order 256 with $d = 5$ (7043 fail F2, 21 610 F3); $r = 2$, $|G| = 2^{11}$: all 7 111 878 groups of order 512 with $d = 5$ (1049 fail F2, 7 110 829 F3'). Logs: `work/p16_14/r256log*`, `r512log_*`, `r128log`, `r256r2log_*`, `r512r5log_*`. $\square$

B.2 Problem 20.37

Let $L(G) = \{|A| : G = AB \text{ for some } A, B \subseteq G\}$. (a) $G = AB$ with $|A||B| = |G|$ iff all products ab are distinct iff $A^{-1}A \cap BB^{-1} = \{1\}$; (b) $L(G)$ is closed under $a \mapsto |G|/a$ (as $G = B^{-1}A^{-1}$); (c) for $H \leq G$ with left transversal T and right transversal T' , $G = TH = HT'$, so $L(G) \supseteq L(H) \cup |G : H|L(H)$. The chain-achievable set $L'(G)$ is the closure of $\{1, |G|\}$ under (b) and (c) over all subgroups (maximal subgroups recursively suffice).

Lemma B.4. *Let $H \leq G$, $|H| = h$, and $x_1 = 1, x_2, \dots, x_k \in G$ with distinct right cosets Hx_i ; put $A = \bigcup Hx_i$. If $B \subseteq G$ is such that the k -sets $\{Hx_1b, \dots, Hx_kb\}$ ($b \in B$) partition $H \setminus G$, then $G = AB$.*

Proof. $AB = \bigcup_{i,b} Hx_ib$ is the union of all right cosets, each once, so $AB = G$ and $|A||B| = |G|$. $\square$

Theorem B.5. *Let $H \leq K \leq G$ with $|K : H|$ even and $K = \langle H, x \rangle$ for some $x \in K$ (necessarily $x \notin H$). Then there is $B \subseteq K$ with $|B| = |K|/(2|H|)$ and $K = (H \cup Hx)B$; in particular $2|H| \in L(K) \subseteq L(G)$.*

Proof. Let Γ be the simple graph with vertex set $H \setminus K$ and an edge $\{Hb, Hxb\}$ for every $b \in K$ with $Hb \neq Hxb$. K acts on $H \setminus K$ by right multiplication, mapping the edge $\{Hb, Hxb\}$ to $\{Hbg, Hxbg\}$, so K acts on Γ by automorphisms, transitively on vertices. The neighbours of Hb are the cosets $Hx^{\pm 1}hb$ ($h \in H$), so the vertices reachable from Hb are the Hwb with

$w \in \langle H, x \rangle = K$: Γ is connected. By the Gallai–Edmonds structure theorem [6], the set D of vertices missed by some maximum matching is $\text{Aut}(\Gamma)$ -invariant, hence $\emptyset$ or all of $V(\Gamma)$; in the latter case the connected graph Γ would be factor-critical and have an odd number of vertices, contradicting $|K : H|$ even. So $D = \emptyset$ and Γ has a perfect matching M . Choosing b_e with $e = \{Hb_e, Hxb_e\}$ for each $e \in M$ and $B = \{b_e\}$, the 2-sets $\{Hb, Hxb\}$ ($b \in B$) partition $H \setminus K$, and Lemma B.4 applied in K gives $K = (H \cup Hx)B$. $\square$

Lemma B.6 (two-sided construction). *Let $H, K \leq G$ with all (H, K) -double cosets regular ($|HgK| = |H||K|$ for all g), $\Omega = H \setminus G/K$, $X = \{x_1, \dots, x_k\}$ and $Y = \{y_1, \dots, y_l\} \subseteq G$ such that the k -sets $\{Hx_i y_j K : i\}$ ($j = 1, \dots, l$) partition Ω . Then $G = AB$ with $A = HX$ and $B = YK$, $|A| = k|H|$, $|B| = l|K|$.*

Proof. $AB = \bigcup_{i,j} Hx_i y_j K$ is the union of all double cosets, each once; by regularity $|AB| = kl|H||K| = |A||B|$. $\square$

In the computations, X is chosen at random and Y is found by an exact-cover search on Ω (a dancing-links solver with node limits, repeated with new X); all found pairs (A, B) are verified by direct multiplication in GAP (`work/p20_37/verify_all12.g`: 286 factorisations, 0 failures).

B.3 Problem 20.21

Let $G, K \cong L$ be as in the problem and $I = K \cap L$. Then G/I embeds in $C_{12} \times A_4$, $|K/I| = |L/I| \in \{4, 12\}$, and the case 12 leads, via an isomorphism $\varphi : L \rightarrow K$, to a smaller configuration of the same kind (so an example of minimal order is not of this type, and every example contains one of the case-4 type). In the case 4, $K/I \cong V_4$, $L/I \cong C_4$, $G/I \cong C_4 \times A_4$, and an element s mapping to a 3-cycle acts on K/I as a 3-cycle. With $I' = \varphi(I) \leq K$ one has $K/I' \cong C_4$ and $I' \cong I$. Hence any example satisfies the *local condition* (L): a group M with normal subgroups $I, I' \cong I$, $M/I \cong V_4$, $M/I' \cong C_4$, and an automorphism α with $\alpha(I) = I$ permuting the three non-trivial cosets of I cyclically.

Lemma B.7. *No abelian group satisfies (L).*

Proof. Induction on $|M|$. $\Omega_1(M)$ is characteristic, so its image in $M/I \cong V_4$ is α -invariant, hence 1 or V_4 . As M is abelian, $\Omega_1(M) \cap I = \Omega_1(I)$ and $\Omega_1(M) \cap I' = \Omega_1(I')$, so $|\Omega_1(M) \cap I| = |\Omega_1(M) \cap I'|$. If $\Omega_1(M)I = M$ then $|\Omega_1(M) \cap I'| = |\Omega_1(M)|/4$, so $\Omega_1(M)I' = M$ and M/I' would be elementary abelian, contradicting $M/I' \cong C_4$. Hence $\Omega_1(M) \leq I$, so $\Omega_1(M) = \Omega_1(I)$; as $|\Omega_1(I')| = |\Omega_1(I)|$ and $\Omega_1(I') \leq \Omega_1(M)$, also $\Omega_1(M) = \Omega_1(I') \leq I'$. Now $(M/\Omega_1(M); I/\Omega_1(I), I'/\Omega_1(I'), \bar{\alpha})$ is a smaller configuration (L) ($I/\Omega_1(I) \cong I'/\Omega_1(I')$ because $I \cong I'$). $\square$

Lemma B.8. *No configuration (L) has $II' \neq M$ (equivalently $|I : I \cap I'| = 2$).*

Proof. Induction on $|M|$; write $N_2(X)$ for the number of involutions in X . Since α permutes the cosets aI, bI, cI cyclically, $N_2(aI) = N_2(bI) = N_2(cI) =: c_2$ and $N_2(M) = N_2(I) + 3c_2$. With $\bar{u}$ generating M/I' , the cosets uI', u^3I' contain no involutions, so $N_2(M) = N_2(I') + N_2(u^2I')$, and $N_2(I) = N_2(I')$ gives $N_2(u^2I') = 3c_2$. Now II' has index 2 in M ; II'/I' is the subgroup of order 2 of M/I' , so $II' = I' \cup u^2I'$, and $II' = I \cup aI$ for one non-trivial coset. Counting involutions in II' in two ways gives $N_2(I) + c_2 = N_2(I') + 3c_2$, so $c_2 = 0$: all involutions lie in I and in I' . Then $W = \Omega_1(M) = \Omega_1(I) = \Omega_1(I') \leq I \cap I'$, $W \neq 1$, and $(M/W; I/W, I'/W, \bar{\alpha})$ is a smaller configuration (L) with $|I/W : (I \cap I')/W| = 2$. $\square$

Consequently an example of minimal order must have $K = I \cdot \varphi(I)$. Computationally (`work/p20_21`): no group of order ≤ 2000 in an order divisible by 12 with $\leq 30\,000$ groups has the required normal subgroups; and no group of order ≤ 256 , of order 512 and rank ≤ 4 , or of order 260–1000 divisible by 4 (except 512 of rank ≥ 5 and 768) satisfies (L). Since an example

of order ≤ 3072 would contain a case-4 example whose K has order ≤ 256 , no example of order ≤ 3072 exists (this covers the orders skipped by the direct search), and an example, if it exists, has $|G| \geq 6144$.

B.4 Problem 19.56

Condition (M): $|ab| \geq |a||b|$ for all $\star$ -commutators a, b of coprime orders. (M) is inherited by subgroups. Facts: (b) in a dihedral group $\langle b, t \rangle$ with b of odd prime order s and t an involution inverting b , $[b, t] = b^{-2}$ generates $\langle b \rangle$ and is a $\star$ -commutator; (c) in $A_4 = V_4 \rtimes \langle x \rangle$, the commutators $[x, y]$ ($y \in V_4 \setminus 1$) run over $V_4 \setminus 1$, so every involution of A_4 is a $\star$ -commutator; (d) in $V \rtimes \langle y \rangle$ with V elementary abelian and y of prime order acting fixed-point-freely, every non-trivial element of V is $[v, y]$ for some v .

Theorem B.9. *Every minimal simple group S contains $\star$ -commutators a (an involution) and b (of odd prime order s) with a inverting b , so $|ab| = 2 < 2s = |a||b|$ and S violates (M). Hence every group with a minimal simple subgroup violates (M); every counterexample to 19.56 contains a minimal non-soluble subgroup which is again a counterexample and has non-trivial Frattini subgroup; and a counterexample of minimal order is minimal non-soluble with non-trivial Frattini subgroup.*

Proof. By Thompson [7], S is $\text{PSL}(2, 2^f)$ ($f \geq 2$ prime), $\text{PSL}(2, 3^f)$ (f odd prime), $\text{PSL}(2, p)$ ($p > 3$, $p \equiv \pm 2 \pmod{5}$), $\text{Sz}(2^f)$ (f odd prime) or $\text{PSL}(3, 3)$. For $S = \text{PSL}(2, 2^f)$: the Borel subgroup is $V \rtimes C_{q-1}$ with C_{q-1} fixed-point-free on $V \setminus 1$, so by (d) every involution (one class) is a $\star$ -commutator; S has a dihedral subgroup of order $2(q \pm 1)$ whose cyclic part contains an element b of order 3, inverted by an involution t ; by (b) $b' := b^{-2}$ is a $\star$ -commutator and $a = t$ inverts it. For $S = \text{PSL}(2, q)$, q odd ≥ 5 : $S \supseteq A_4$, so involutions are $\star$ -commutators by (c); some odd prime s divides $(q-1)/2$ or $(q+1)/2$ (both cannot be powers of 2 for $q > 3$), and the dihedral subgroup of order $q \mp 1$ provides b of order s and an inverting involution. For $S = \text{Sz}(q)$: the involutions are the non-trivial elements of $Z(P)$ (P a Sylow 2-subgroup) on which C_{q-1} acts fixed-point-freely, so (d) applies; the normaliser of C_{q-1} is dihedral and $q-1$ has an odd prime factor. For $\text{PSL}(3, 3)$ the pair was found by direct computation (it contains S_4 and D_{12}). The remaining assertions follow since (M) passes to subgroups, every non-soluble group contains a minimal non-soluble subgroup, and a minimal non-soluble group with trivial Frattini subgroup is minimal simple [7]. $\square$

The 208 non-soluble groups of order ≤ 2000 (in orders with ≤ 5000 groups) and all 443 perfect groups of order $\leq 10^5$ violate (M) (work/p19_56). The lifting of the configuration through a non-trivial Frattini subgroup is the remaining obstacle; the agent's partial observations are in the report.

B.5 Problem 21.26

Proposition B.10. *If $|G|$ has exactly two prime divisors p, q , with Sylow subgroups P and Q (so $G = PQ = QP$), there is x with $P \cap P^x$ and $Q \cap Q^x$ both inclusion-minimal.*

Proof. For $a \in P$, $P \cap P^{ga} = (P \cap P^g)^a$, and conjugation by a permutes the family $\{P \cap P^g\}$, so minimality is invariant under $g \mapsto ga$. Choose g with $P \cap P^g$ minimal and write $g = ba$ ($b \in Q$, $a \in P$); then $P \cap P^b$ is minimal. Likewise choose h with $Q \cap Q^h$ minimal, $h = a'b'$ ($a' \in P$, $b' \in Q$); then $Q \cap Q^{a'}$ is minimal. For $x = ba'$: $P \cap P^x = (P \cap P^b)^{a'}$ and $Q \cap Q^x = Q \cap (Q^b)^{a'} = Q \cap Q^{a'}$, both minimal. $\square$

B.6 Problem 21.111

Theorem B.11. *For $q = 3^n$, $n \geq 2$, $S = \text{PSL}(2, q)$ is a τ -group: for n odd the involutions of S , for n even the outer (diagonal) involutions of $\text{PGL}(2, q)$ are τ -automorphisms.*

Proof. For an involution $x \in G = \langle x, \text{Inn } S \rangle$ and $s \in S$, $\langle x, x^s \rangle$ is dihedral of order $2 \cdot o(xx^s)$, so x is τ iff no product of two conjugates of x has order divisible by 3. Elements of order divisible by 3 are unipotent (orders in S are 3 and divisors of $(q \pm 1)/2$, and $3 \nmid (q \pm 1)/2$). If $u = x_1 x_2$ with involutions x_1, x_2 conjugate in G then x_1 inverts u , so $x_1 \in N_G(\langle u \rangle)$. For n odd, $q \equiv 3 \pmod{4}$ and $N_S(\langle u \rangle)$ is the Borel subgroup of odd order $q(q-1)/2$, so no involution of S inverts u . For n even the involutions of $\text{PGL}(2, q)$ inverting u lie in its Borel subgroup, whose involutions are conjugate to the image of $\text{diag}(-1, 1)$, which lies in $\text{PSL}(2, q)$ since -1 is a square ($q \equiv 1 \pmod{4}$); so the outer involutions never invert a unipotent element. $\square$

C Commit timeline

All 94 commits made during the 48-hour run (date and time in UTC, month-day hour:minute, and the commit message); the commits adding this paper are not listed.

Time	Commit message
09-13 08:04	Initial setup: README, progress log, gitignore
09-13 08:14	Plan, first search scripts (21.113, 21.29, 21.99, 21.52)
09-13 08:24	More search scripts (21.26, 21.115, 21.113 Brauer) and logs
09-13 08:52	21.32: decidability proof draft; more sweeps
09-13 09:00	Progress log; new searches (21.111, 21.25, 21.27, 21.135)
09-13 09:13	16.60 solved (twisted FS reduction); new sweeps for 14.44/14.74, 15.2, 16.14, 11.18, 13.19, 16.95, 11.17, 11.8
09-13 09:17	plan update with triage results
09-13 13:03	verification report + progress
09-13 13:03	16.14 partial result notes
09-13 13:15	21.111 partial report; more sweeps
09-13 13:26	2.78: finiteness of soluble and absolutely simple numbers (report)
09-13 13:27	2.78 report: clarify Lemma C
09-13 13:27	progress
09-13 13:36	more sweeps (20.21, 20.86, 20.30, 19.30)
09-13 13:44	progress; 18.46 data
09-13 13:50	18.46: D8 needs $2 G ^2$ (verification running)
09-13 13:52	18.46 solved: bound $2 G ^2$ is sharp (D8)
09-13 13:53	plan status
09-13 13:57	verification updates
09-13 14:09	18.46 report: 14 groups of order 128; verification updates
09-13 14:09	18.46 final; progress
09-13 14:15	17.1 exponent-3 negative; progress
09-13 14:18	13.19 counterexample draft
09-13 14:19	results index
09-13 14:21	13.19 solved: counterexample of order 512 verified
09-13 14:24	verification updates; stopped low-value sweeps
09-13 14:26	progress
09-13 14:27	2.78: data remark
09-13 14:28	19.12, 20.100 verifications
09-13 14:32	18.117: sporadic verification report (partial)
09-13 14:34	progress: 18.117, 19.12, 20.100, 20.71
09-13 14:38	13.19: general theorem (every p-group occurs as Q/H), Mobius lemma
09-13 14:38	13.19: fix order of tagged D8 example
09-13 14:39	reports: 21.2, 20.71, 17.1, 18.117 updates
09-13 14:41	20.21 partial notes
09-13 14:42	16.60 report: clean remark
09-13 14:44	verifications: 15.3, 19.33
09-13 14:51	21.26: small-group sweep ≤ 600 (no counterexample), double-coset checker, launch 601-2000 sweep
09-13 14:55	18.119 targeted sweep done; 21.26 heavy-order sweeps launched

Time	Commit message
09-13 14:56	fix timestamps in progress/plan (real clock from git log)
09-13 14:56	fix timestamp in results index
09-13 14:58	18.120 verification recorded
09-13 15:13	20.71: n=11 exhaustive (no example); 16.14 order 2^{11} excluded, 2^{12} search launched
09-13 15:15	verifications: 21.26 (≤ 2000 , big groups), 18.20 library tables
09-13 15:18	16.14: minimal counterexample structure (Theorem B), cohomological exclusion of order 2^{11} ($r=3$), runs for $r=2$ and 2^{12}
09-13 15:22	16.14: order 2^9 ($r=2$) excluded; killgap fix
09-13 15:28	16.14: order 2^{10} fully excluded
09-13 15:31	results index and progress log update
09-13 15:32	20.21: correct the local condition (3-power order automorphisms), rerun
09-13 15:33	20.21: rigorous proof that abelian M cannot satisfy the local condition
09-13 15:40	16.14: order 2^{12} ($r=3$) excluded; 20.21 corrected search to 252 clean; 21.26 order 1920 done
09-13 15:42	21.26: heavy orders done
09-13 15:50	new sweeps: 20.115, 20.78, 19.108, 20.49, 20.122; 20.37 chain analysis; 17.1 p=5 done
09-13 15:54	20.37: chain-achievability analysis, coset-union construction, first factorizations (PSL(2,8), PSL(2,13), PSL(2,17), PSL(2,19))
09-13 15:57	20.37: Theorem 1 ($2 H $ always realisable when $ K:H $ even and $K=\langle H, x \rangle$), closure computation
09-13 16:03	19.108 sweep done (no counterexample); 20.122 sweep over all nilpotent subgroups; 20.37 progress
09-13 16:07	20.37: table of sizes left open by Theorem 1; DLX solver
09-13 16:11	progress: 20.37 two-sided construction; sweeps status
09-13 16:12	20.49 sweep done (no counterexample); 16.14 F3 observation
09-13 16:18	20.37: Lemma 2 (two-sided construction), phase-2 queue
09-13 16:22	20.37/20.21/20.122: running searches, logs and solutions so far
09-13 16:26	20.21: Lemma T2 (configurations with $I \cdot I' \neq M$ are impossible)
09-13 16:27	20.21 notes: remark wording
09-13 16:27	20.4 sweep done (no counterexample)
09-13 16:27	results index: 20.21, 20.37
09-13 16:41	20.37: more factorizations; 20.21 family test
09-13 16:57	20.122: verified range recorded (sweep stopped)
09-13 17:46	19.56: theorem (every minimal simple group violates (M)), computations, remarks on lifting
09-13 17:47	results index: 19.56
09-13 17:47	19.56 sweep done
09-13 17:50	20.37 table counts from solution files; 19.56 perfect-group test
09-13 17:50	18.56 partial verification recorded
09-13 17:54	21.26: two-prime case (proposition), notes
09-13 17:54	19.56: perfect groups $\leq 1e5$ all violate; index updates
09-13 19:35	20.21: local condition excluded for all $ K \leq 256$
09-13 19:36	20.37: 240/400 sizes realised; 20.21 rank-4 order-512 search launched
09-13 20:03	16.14: p-cover based filters (ANUPQ), order-512 rank-5 sweep launched; 20.37 phase-3 queue
09-13 20:08	16.14: p-cover filter description
09-14 02:33	16.14: order 2^{11} fully excluded (7.1M quotients), Theorem C updated; progress stop 20.78/18.56 long runs; record
09-14 04:08	21.113 tail run; logs
09-14 04:28	21.113: sweep to 1000 complete
09-14 09:21	20.21: order-512 rank 3/4 coverage recorded
09-14 09:25	20.37: results section with current table (261/400, 280 verified)
09-14 09:25	20.37: correct the list of completely settled groups
09-14 09:41	20.21: order 512 rank ≤ 4 complete (no candidate)
09-14 11:39	20.21 notes: status and structure paragraph cleaned; 18.117 final list

Time	Commit message
09-14 12:20	20.21: local condition excluded for orders 260-1000 (except 512 rank $\geq$ 5, 768)
09-14 12:20	progress log
09-14 14:14	20.37: phase-4 searches; PSL(2,27) size 63 found
09-14 17:18	20.37: phase-4 progress (264/400)
09-15 00:24	final polish of the index and top-level README
09-15 00:25	verifications table: final ranges, no stale running notes
09-15 09:44	final consolidation: 20.37 final table (267/400, 286 verified), index and progress

D The repository

The repository (git, branch `master`; 94 commits during the run, later commits add this paper) has the following layout.

- `README.md` — entry point; `plans/PLAN.md` — the plan and the triage tables.
- `reports/` — one Markdown report per solved or partially solved problem: `21.32_derived_subgroup_decidability.md`, `16.60_twisted_frobenius_schur.md`, `2.78_group_theoretic_numbers.md`, `18.46_square_roots_sharpness.md`, `13.19_subdirect_products_counterexample.md`, `16.14_notes.md`, `20.37_subset_factorizations.md`, `20.21_notes.md`, `19.56_star_commutators.md`, `21.26_notes.md`, `21.111_tau_groups_partial.md`, `18.117_coprime_commutators_sporadic.md`; the index `README.md`; the table `verifications.md`; and the chronological log `PROGRESS.md`.
- `work/p<problem>/` — 57 directories with the scripts (261 GAP files, 14 Python files, 7 C files) and their logs; each report names the scripts that support its claims. Examples: `work/p21_26/check2.g` (the double-coset checker), `work/p16_14/pcover_filter.g` (the 2-covering-group filter), `work/p20_37/fact4.py`, `dlx.c`, `verify_all2.g` (the factorisation search and its independent verification), `work/p20_71/cards.c` (the nauty filter).
- `session/` — the complete transcript of the run (main session and the six sub-agents, JSON lines).
- `docs/` — the notebook (PDF and text); `paper/` — this paper.

The GAP installation and the Python virtual environment are not part of the repository. All computations were run with GAP 4.16.1 as installed in `gap-4.16.1/`; the scripts use only library functions of GAP and its standard packages, and can be re-run with commands of the form `gap -q -b -o 8g work/p21_26/s2_601_650.g`.